Appendix A
Diagrams of Regression Analysis of Basic Design Values for Merchant Ships

Abstract The present appendix is a collection of design diagrams resulting from the statistical/regression analysis of the main dimensions of basic ship design quantities and pertain to various ship types. The main source of processed data is the IHS Fairplay World Shipping Encyclopedia.[1] The data processing and analysis were primarily conducted by diploma thesis students of the National Technical University of Athens and staff of the Ship Design Laboratory of NTUA.

Notes

1. Whenever an asterisk (*) is presented in the legend of the following diagrams, it means that:
   - The shown *outer* boundary curves (purple) represent the 95% *prediction interval*; thus, 95% of the statistical data are within this interval
   - The shown *inner* boundary curves (purple) represent the statistical 95% *confidence interval* of the resulting regression formula
2. A commonly used indicator of the goodness of fit of the shown regression formulas is the R-squared regression coefficient. Given regression formulas with R-squared less than about 0.60 should be used with great caution.

[1] IHS Fairplay World Shipping Encyclopedia is the largest maritime database in the world, evolved from the Lloyd's Register of Ships published since 1764, covering ship characteristics, movements, ownership, casualties, ports, news and research. It provides comprehensive, quarterly updated information concerning ships of 100 GT and above. Key data and features include details of over 116,000 vessels (100 GT and above); ships on-order and under construction; over 124,000 marine-related companies; over 10,000 ports and terminals; maritime atlas with port plans, mooring diagrams and photographs; export facility; distance tables, see, http://www.ihs.com/products/maritime-information/ships/world-shipping-encyclopedia.aspx.

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A. Papanikolaou, *Ship Design*, https://doi.org/10.1007/978-3-032-11960-5

A.1 Bulk Carriers

See Figs. A.1, A.2, A.3, A.4, A.5, A.6, A.7, A.8, A.9, A.10, A.11, A.12, A.13, A.14, A.15, A.16, A.17, A.18, A.19, A.20 and A.21.

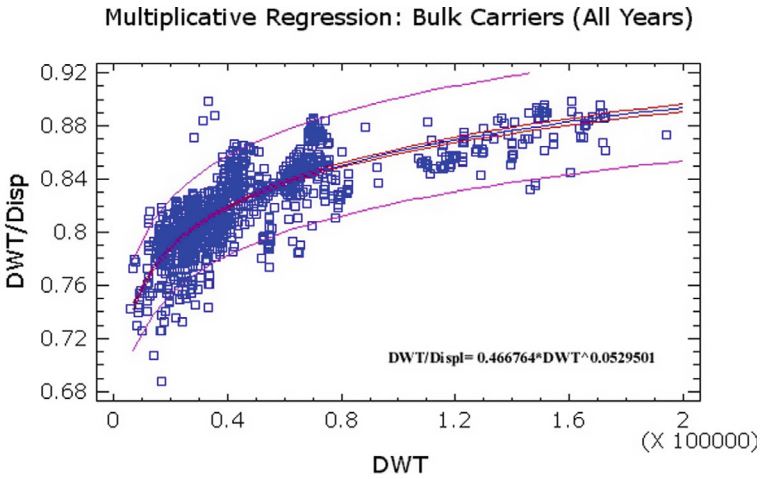


Fig. A.1 Regression analysis of ratio (DWT/displacement) versus DWT [tons] for Bulk Carriers* (Kalokairinos et al. 2000–2005)

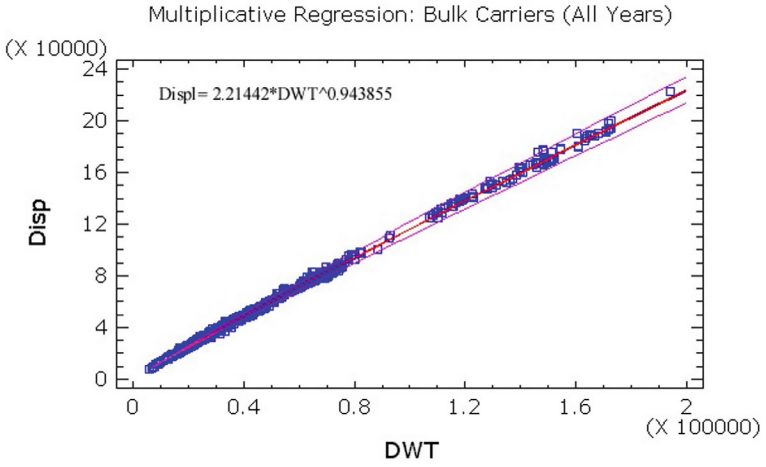


Fig. A.2 Regression analysis of displacement Δ [tons] versus DWT [tons] for Bulk Carriers* (Kalokairinos et al. 2000–2005)

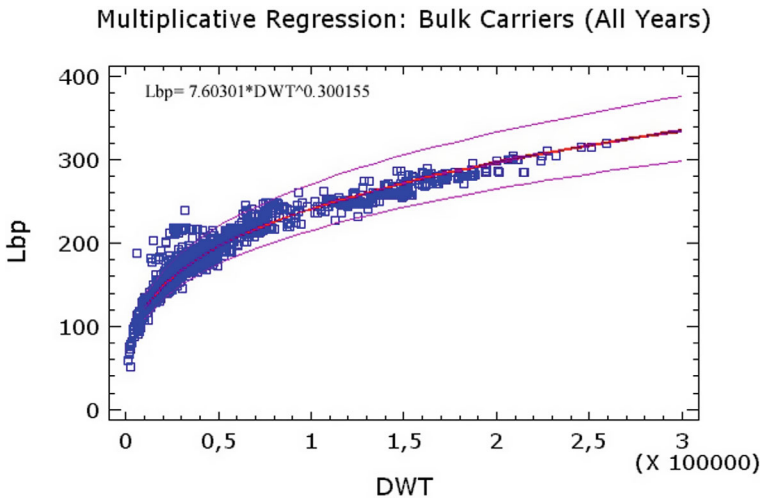


Fig. A.3 Regression analysis of length L_{BP} [m] versus DWT [tons] for Bulk Carriers* (Kalokairinos et al. 2000–2005)

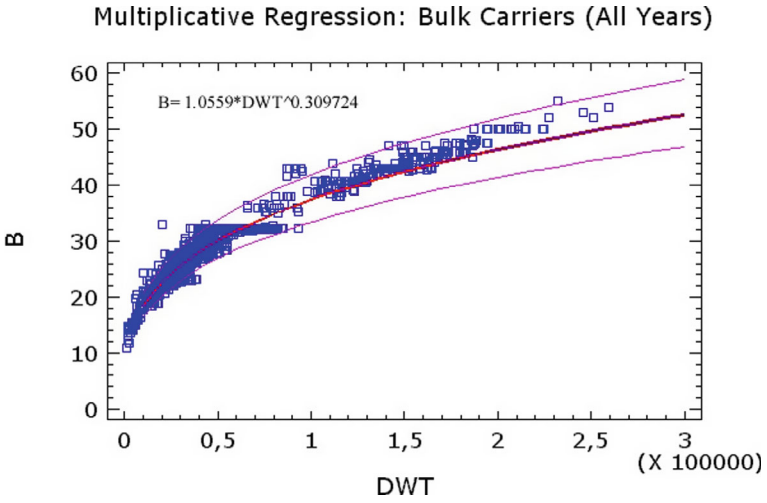


Fig. A.4 Regression analysis of beam B [m] versus DWT [tons] for Bulk Carriers* (Kalokairinos et al. 2000–2005)

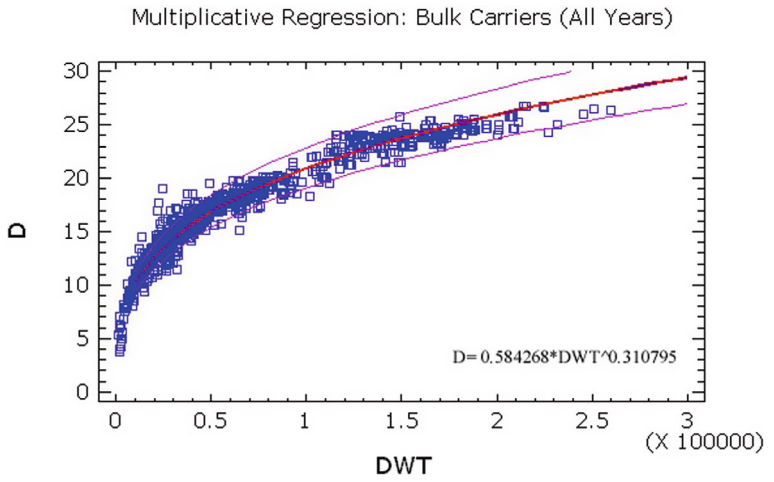


Fig. A.5 Regression analysis of side depth D [m] versus DWT [tons] for Bulk Carriers* (Kalokairinos et al. 2000–2005)

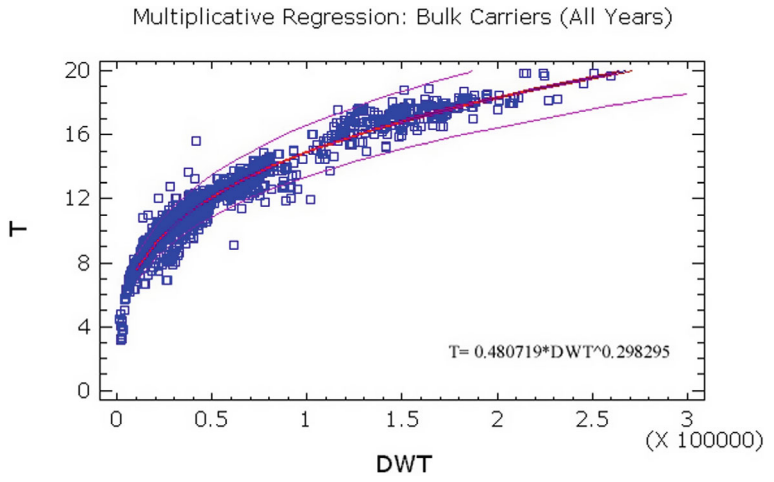


Fig. A.6 Regression analysis of draft T [m] versus DWT [tons] for Bulk Carriers* (Kalokairinos et al. 2000–2005)

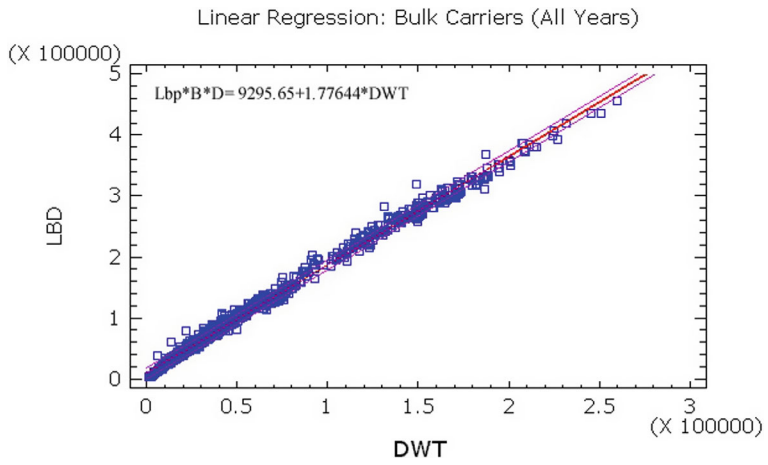


Fig. A.7 Regression analysis of the product (L × B × D) [m^3] versus DWT [tons] for Bulk Carriers* (Kalokairinos et al. 2000–2005)

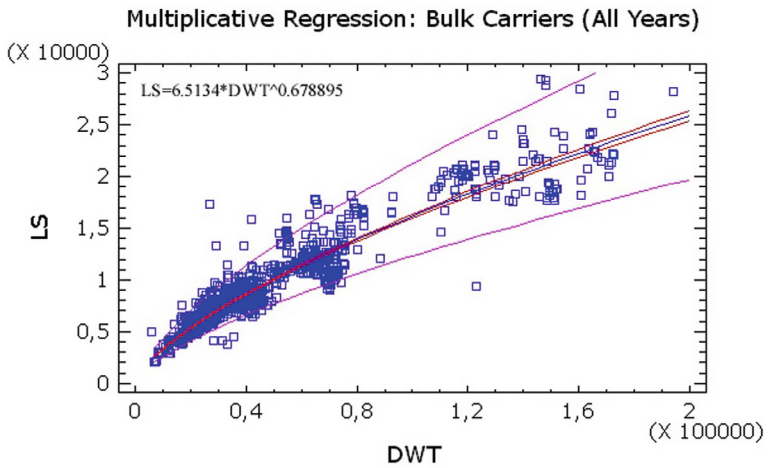


Fig. A.8 Regression analysis of lightship-weight (LS) [tons] versus DWT [tons] for Bulk Carriers* (Kalokairinos et al. 2000–2005)

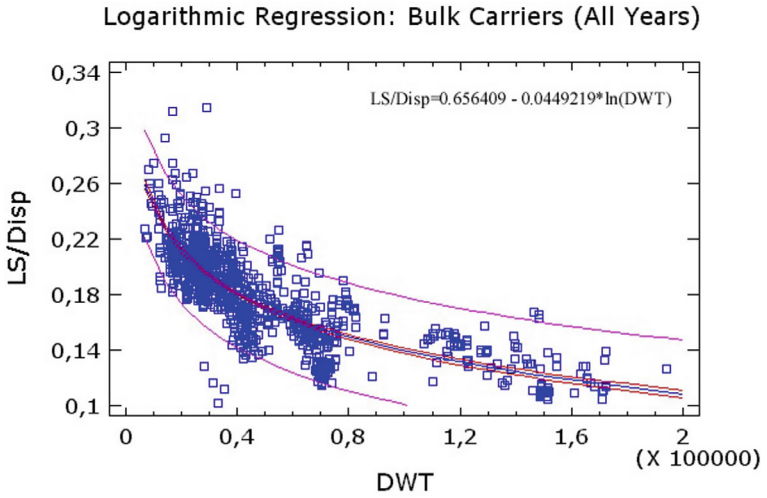


Fig. A.9 Regression analysis of ratio (LS/Δ) versus DWT [tons] for Bulk Carriers* (Kalokairinos et al. 2000–2005)

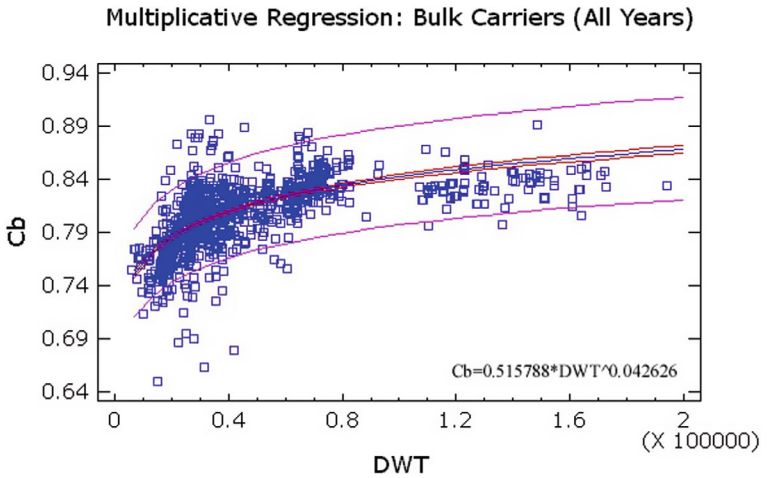


Fig. A.10 Regression analysis of block coefficient C_B versus DWT [tons] for Bulk Carriers* (Kalokairinos et al. 2000–2005)

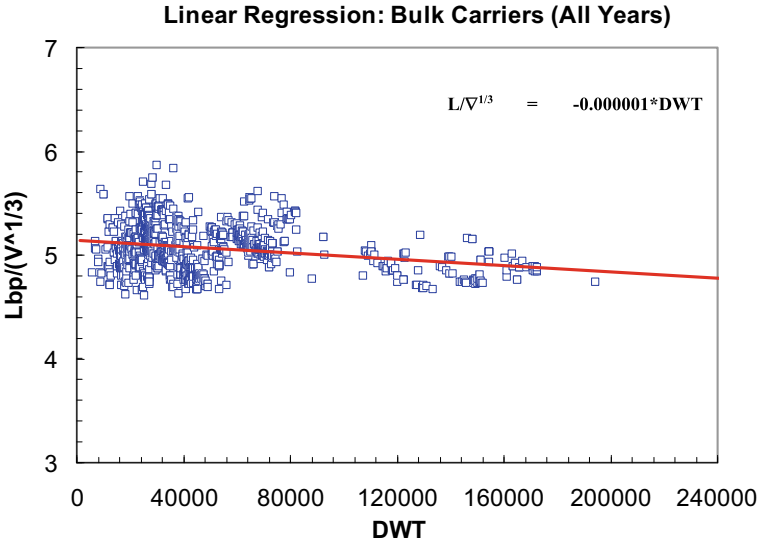


Fig. A.11 Regression analysis of slenderness ratio ($L/\nabla^{1/3}$) versus DWT [tons] for Bulk Carriers (Kalokairinos et al. 2000–2005)

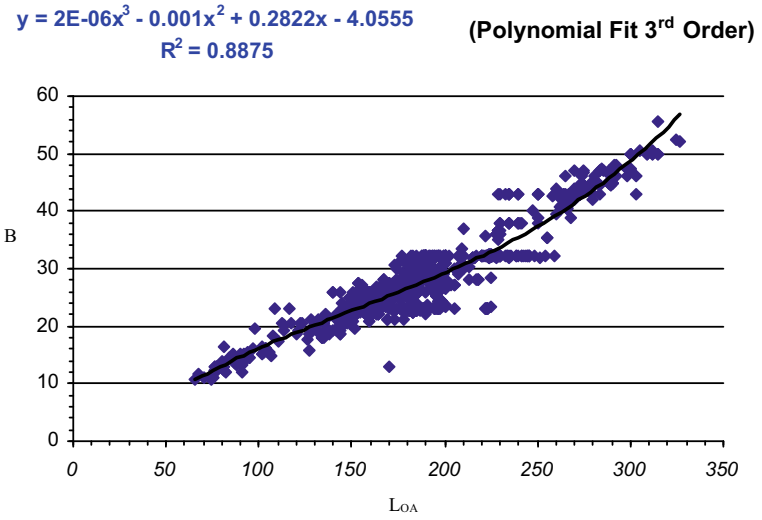


Fig. A.12 Regression analysis of B versus the L_{OA} for Bulk Carriers (Kalokairinos et al. 2000–2005)

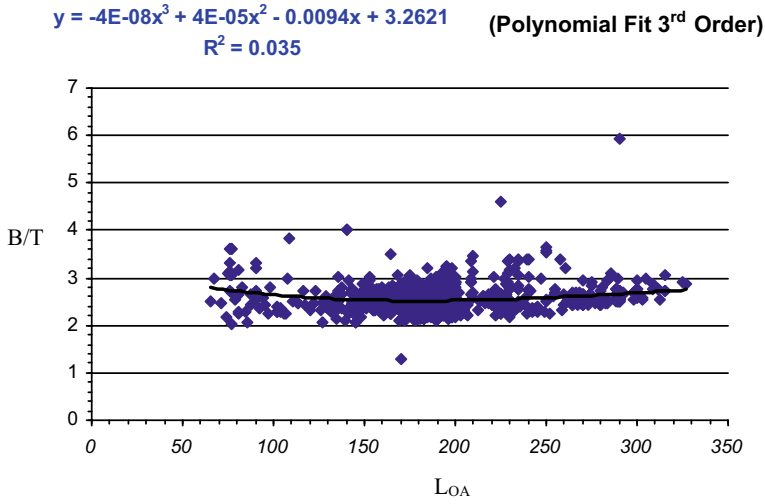


Fig. A.13 Regression analysis of the B/T versus the L_{OA} for Bulk Carriers (Kalokairinos et al. 2000–2005)

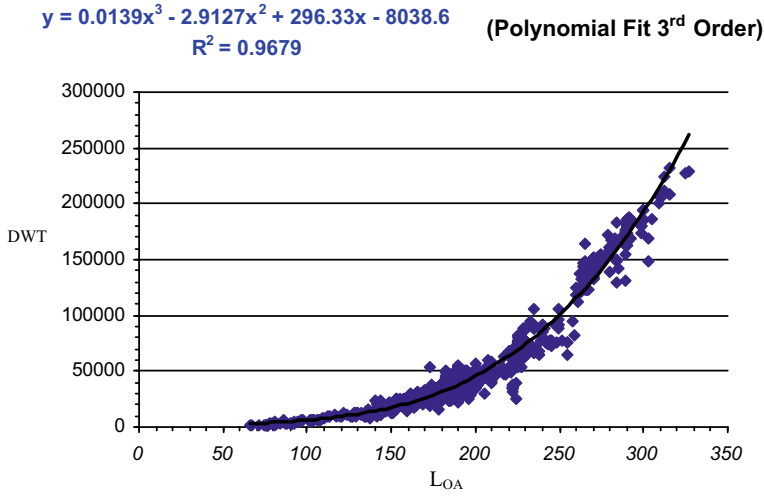


Fig. A.14 Regression analysis of the DWT versus the L_{OA} for Bulk Carriers (Kalokairinos et al. 2000–2005)

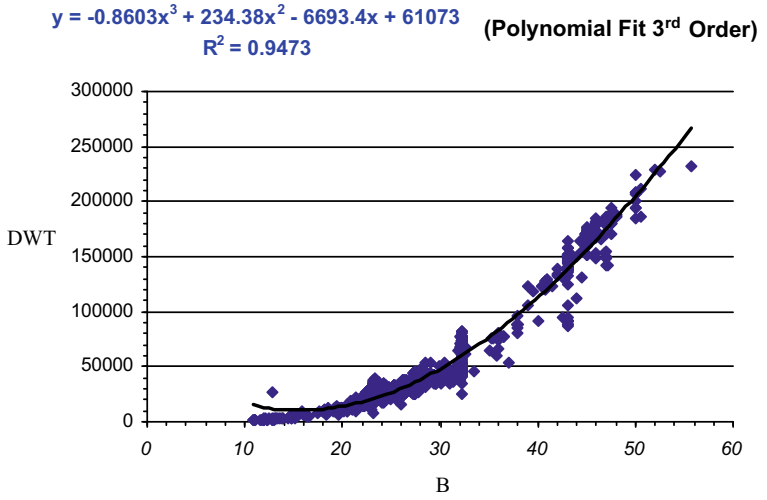


Fig. A.15 Regression analysis of the DWT versus the beam B for Bulk Carriers (Kalokairinos et al. 2000–2005)

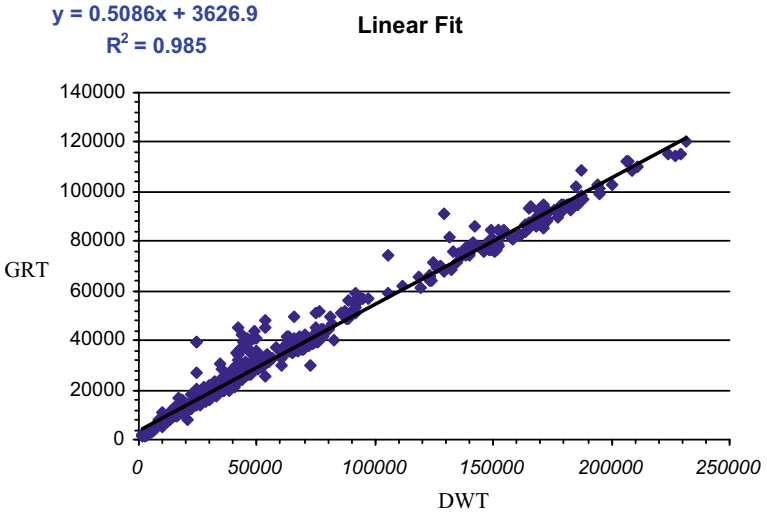


Fig. A.16 Regression analysis of the GRT versus the DWT for Bulk Carriers (Kalokairinos et al. 2000–2005)

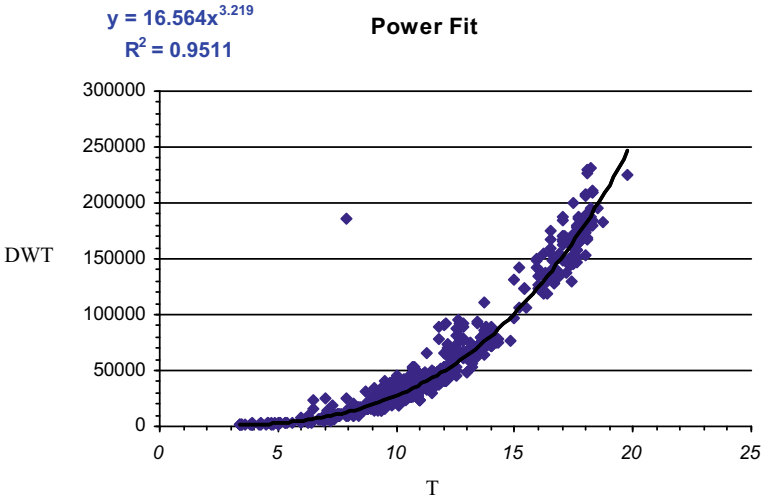


Fig. A.17 Regression analysis of the DWT versus the draft T for Bulk Carriers (Kalokairinos et al. 2000–2005)

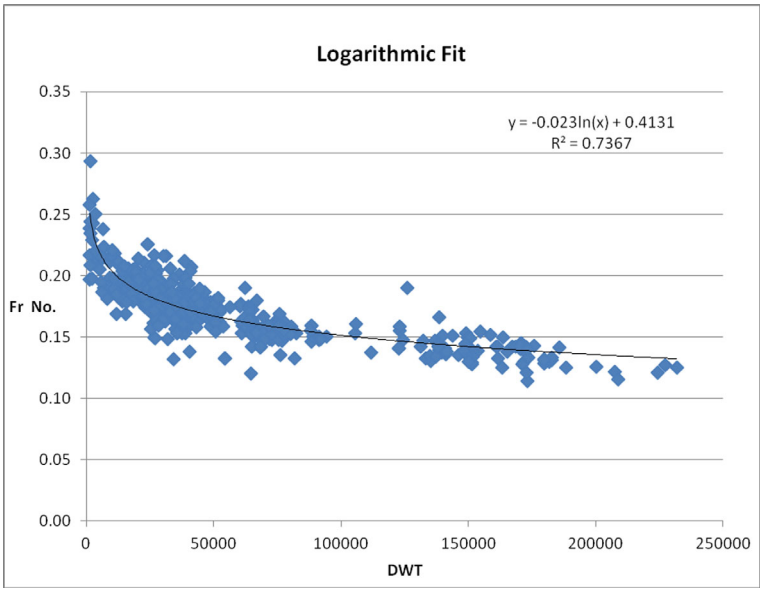


Fig. A.18 Regression analysis of the Froude number versus the DWT for Bulk Carriers (Kalokairinos et al. 2000–2005)

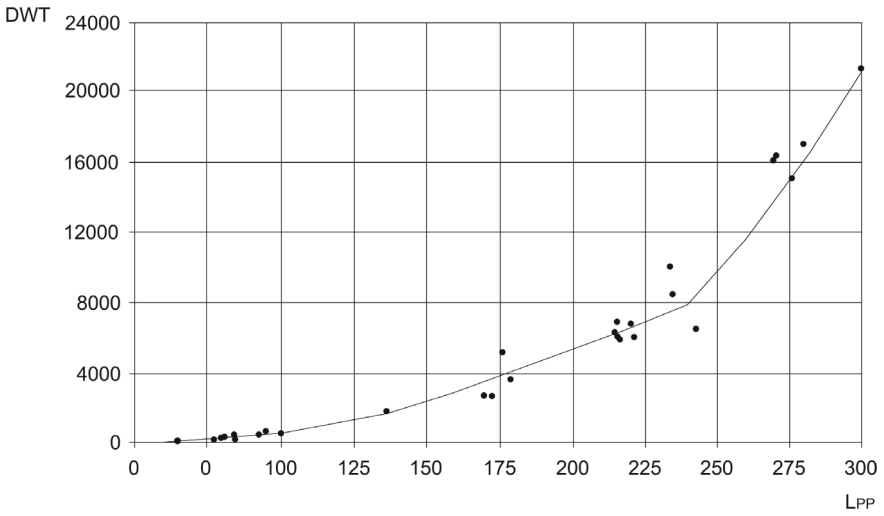


Fig. A.19 Regression analysis of the DWT [tons] versus the length L [m] for bulk carriers according to Kristensen (2000) in Friis et al. (2002)

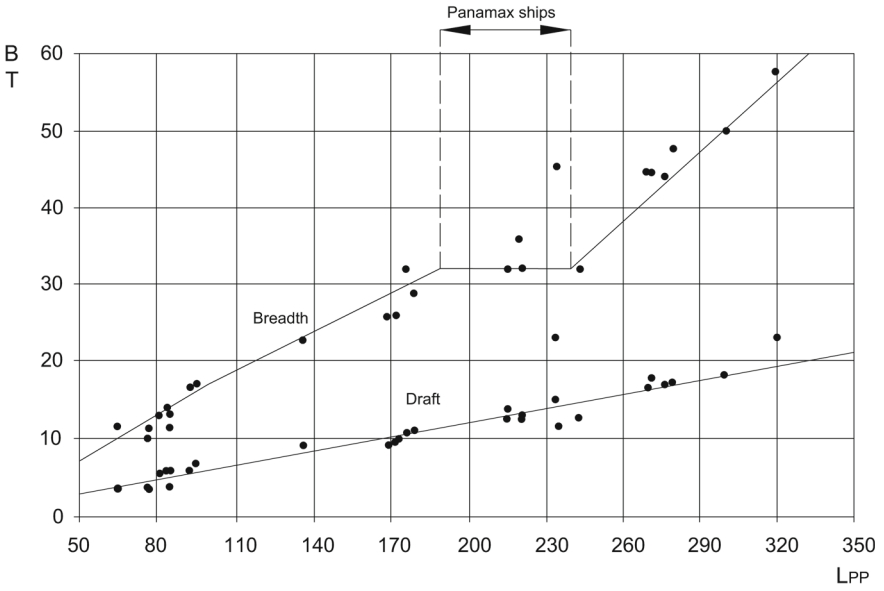


Fig. A.20 Regression analysis of the beam B [m] and the draft T [m] versus the length L [m] for bulk carries according to Kristensen (2000) in Friis et al. (2002)

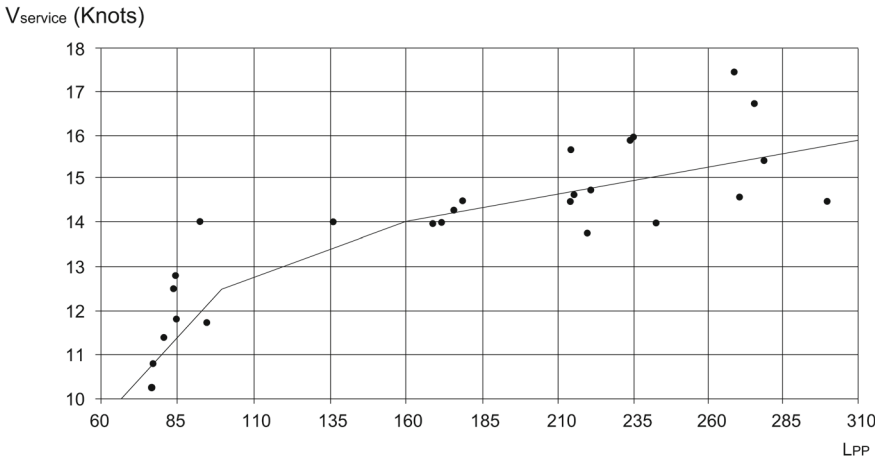


Fig. A.21 Regression analysis of the service speed Vs [knots] versus the length L [m] for bulk carriers according to Kristensen (2000) in Friis et al. (2002)

A.2 OBO Carriers

See Figs. A.22, A.23, A.24, A.25, A.26, A.27 and A.28.

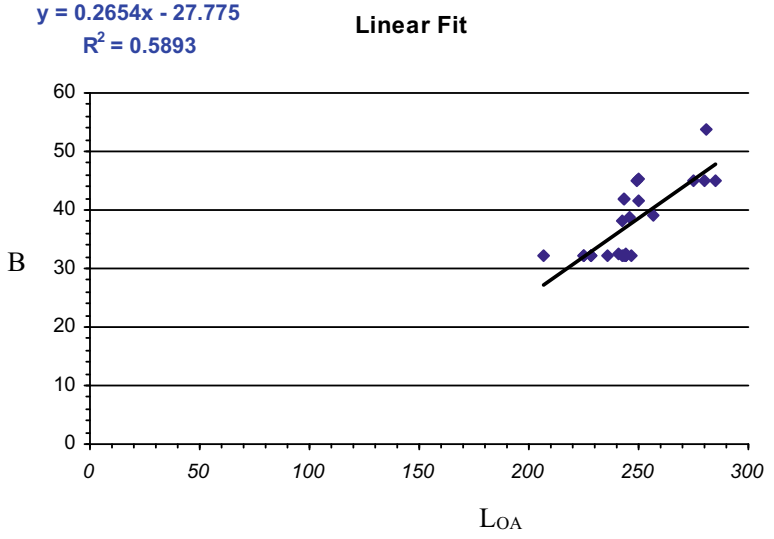


Fig. A.22 Regression analysis of beam B [m] versus length L_{OA} [m] for OBO Carriers (Kalokairinos et al. 2000–2005)

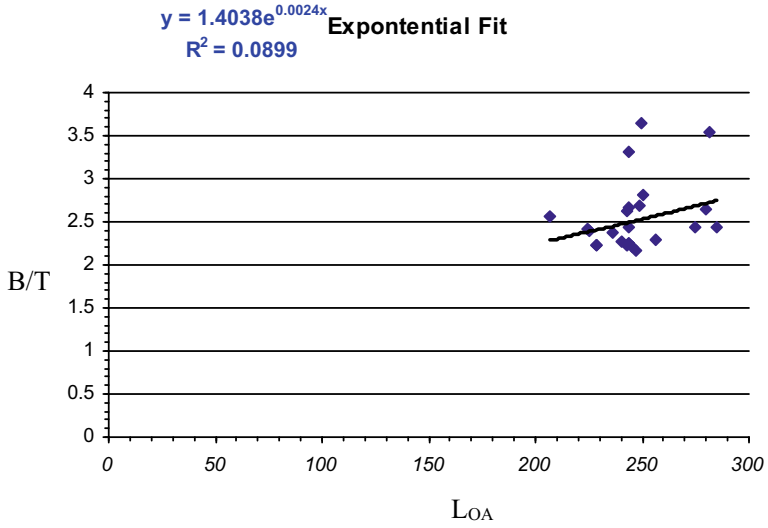


Fig. A.23 Regression analysis of ratio B/T versus length L_{OA} [m] for OBO Carriers (Kalokairinos et al. 2000–2005)

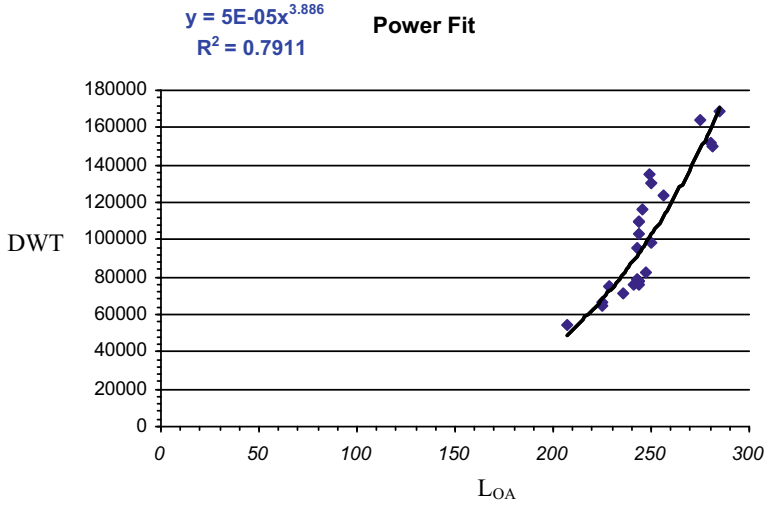


Fig. A.24 Regression analysis of the DWT [tons] versus length L_{OA} [m] for OBO Carriers (Kalokairinos et al. 2000–2005)

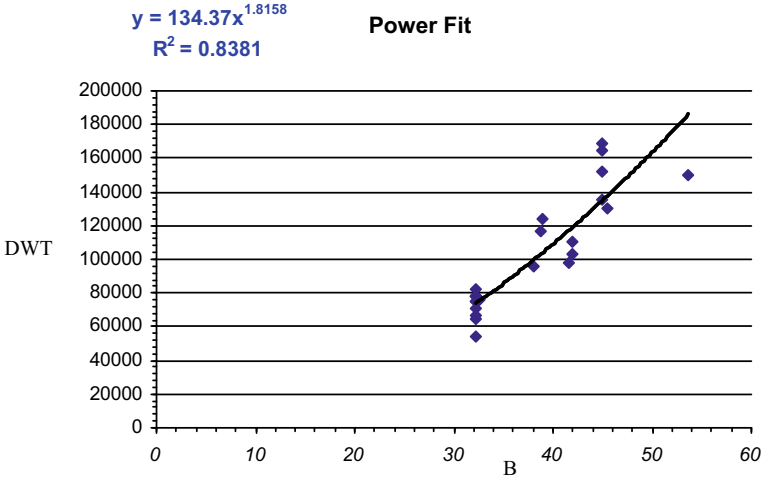


Fig. A.25 Regression analysis of DWT [tons] versus beam B [m] for OBO Carriers (Kalokairinos et al. 2000–2005)

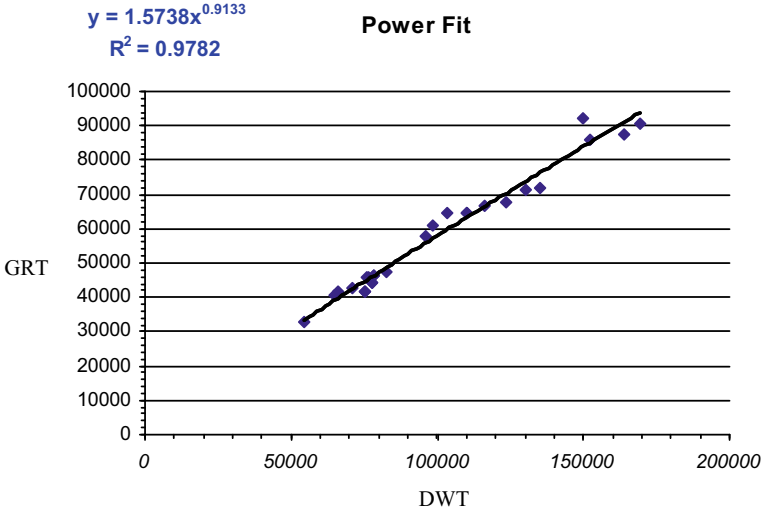


Fig. A.26 Regression analysis of GRT [RT] versus DWT [tons] for OBO Carriers (Kalokairinos et al. 2000–2005)

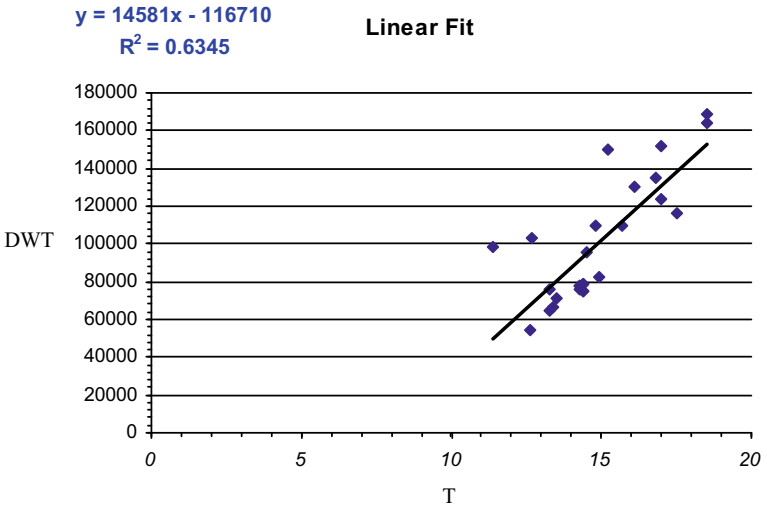


Fig. A.27 Regression analysis of DWT [tons] versus draft T [m] for OBO Carriers (Kalokairinos et al. 2000–2005)

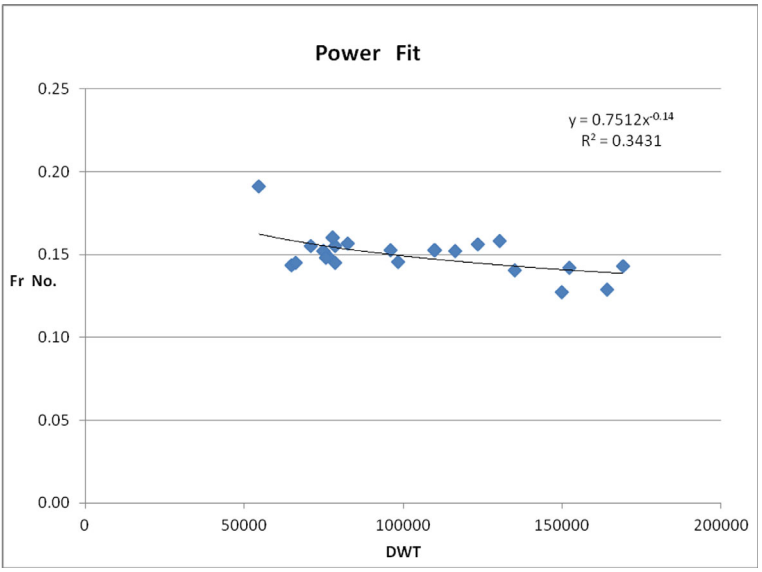


Fig. A.28 Regression analysis of the Froude no. versus DWT [tons] for OBO Carriers (Kalokairinos et al. 2000–2005)

A.3 Containerships

See Figs. A.29, A.30, A.31, A.32, A.33, A.34, A.35, A.36, A.37, A.38, A.39, A.40, A.41, A.42, A.43, A.44, A.45, A.46, A.47, A.48, A.49 and A.50.

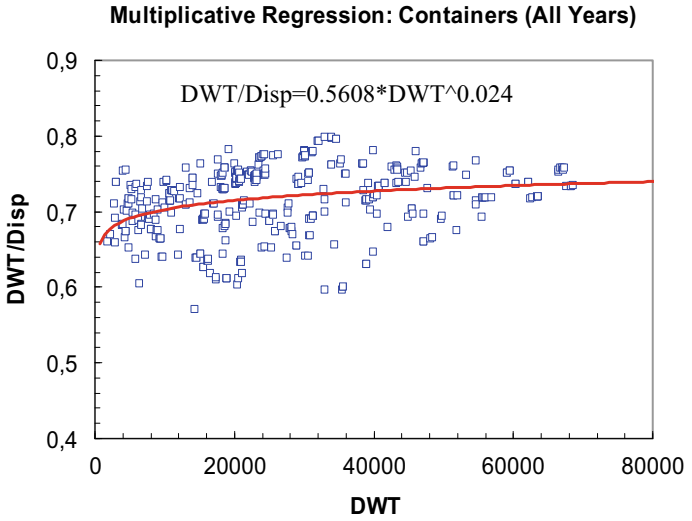


Fig. A.29 Regression analysis of ratio (DWT/Δ) versus DWT [tons] for Containerships (Kalokairinos et al. 2000–2005)

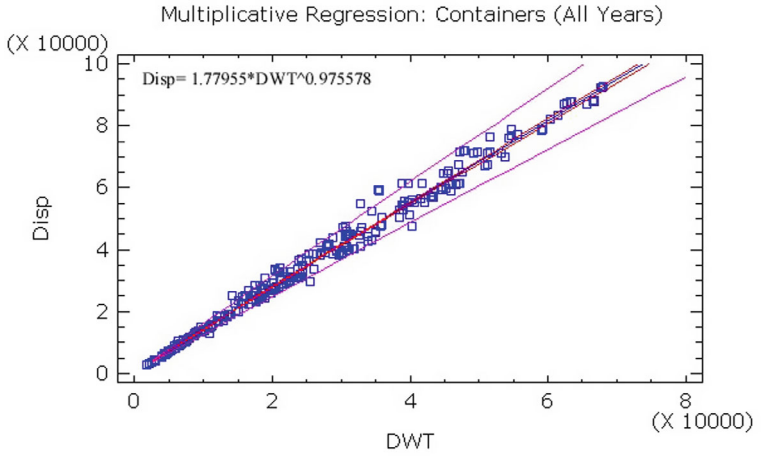


Fig. A.30 Regression analysis of displacement Δ [tons] versus DWT [tons] for Containerships* (Kalokairinos et al. 2000–2005)

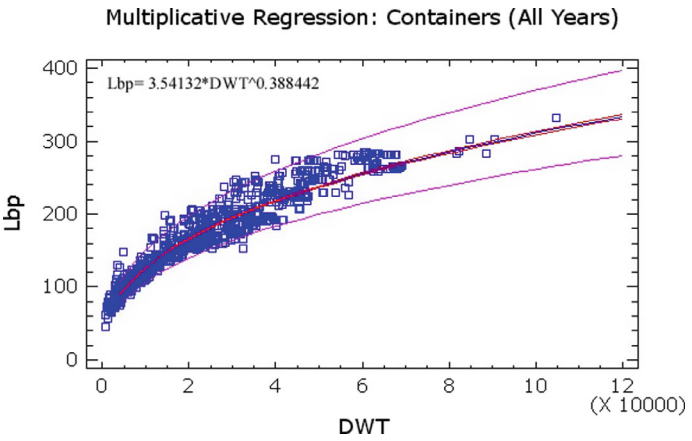


Fig. A.31 Regression analysis of length L_{BP} [m] versus DWT [tons] for Containerships* (Kalokairinos et al. 2000–2005)

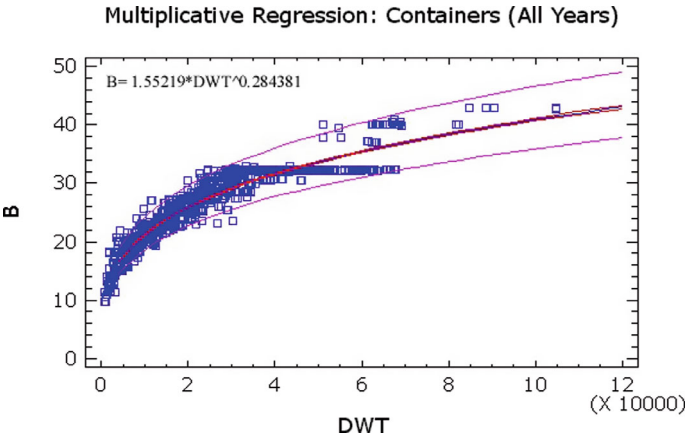


Fig. A.32 Regression analysis of beam B [m] versus DWT [tons] for Containerships* (Kalokairinos et al. 2000–2005)

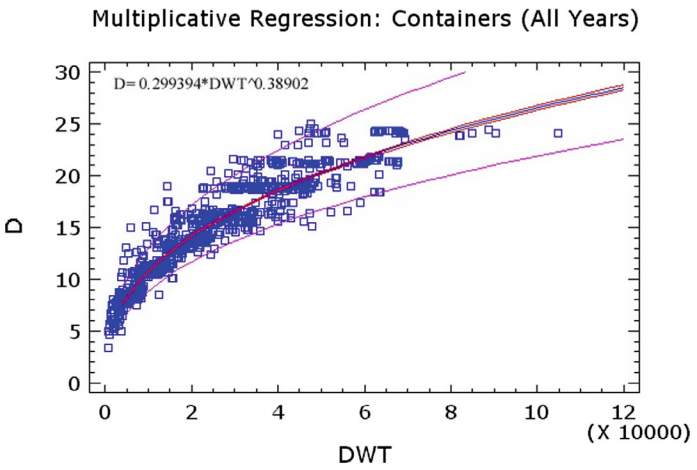


Fig. A.33 Regression analysis of side depth D [m] versus DWT [tons] for Containerships* (Kalokairinos et al. 2000–2005)

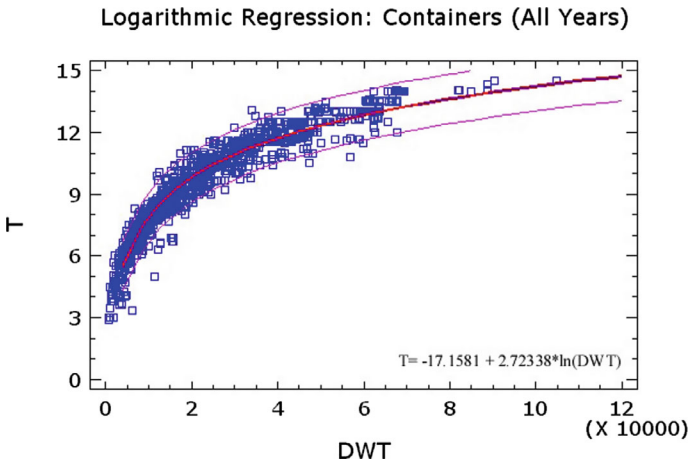


Fig. A.34 Regression analysis of draft T [m] versus DWT [tons] for Containerships* (Kalokairinos et al. 2000–2005)

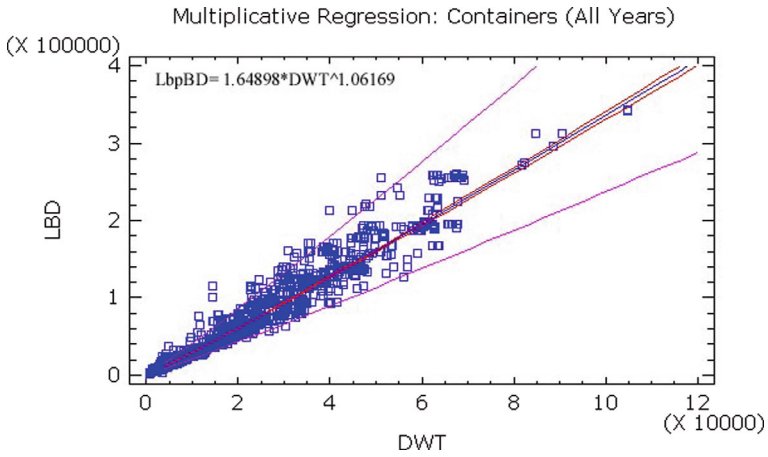


Fig. A.35 Regression analysis of volumetric product (L × B × D) [m^3] versus DWT [tons] for Containerships* (Kalokairinos et al. 2000–2005)

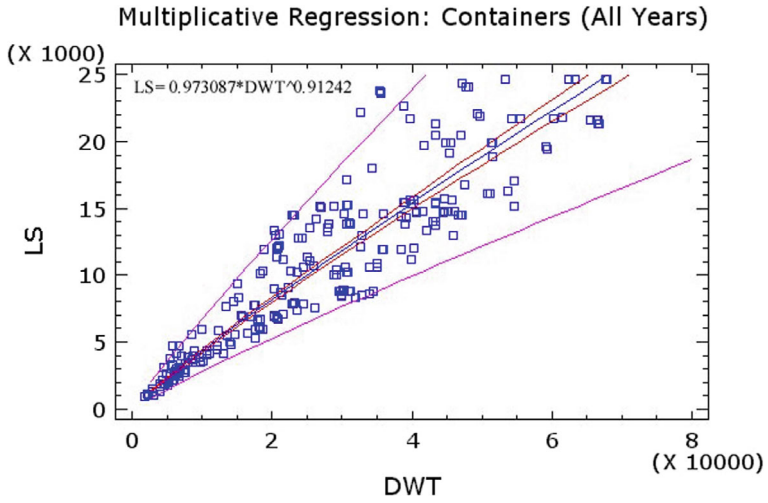


Fig. A.36 Regression analysis of lightship LS [tons] versus DWT [tons] for Containerships* (Kalokairinos et al. 2000–2005)

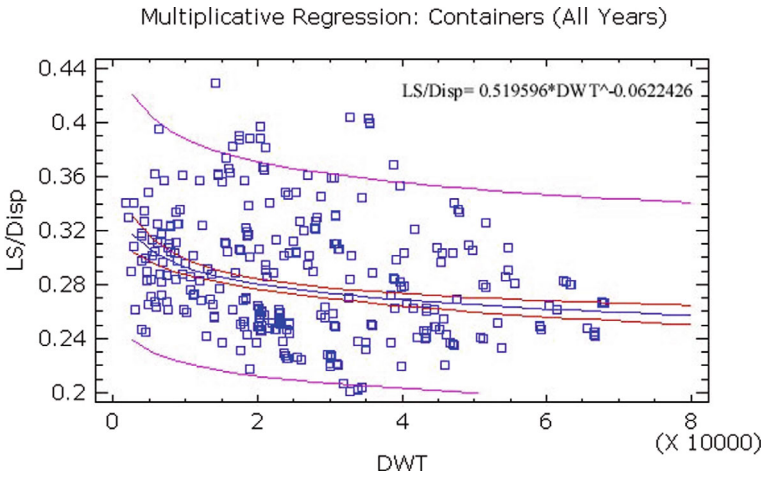


Fig. A.37 Regression analysis of ratio (LS/Δ) versus DWT [tons] for Containerships* (Kalokairinos et al. 2000–2005)

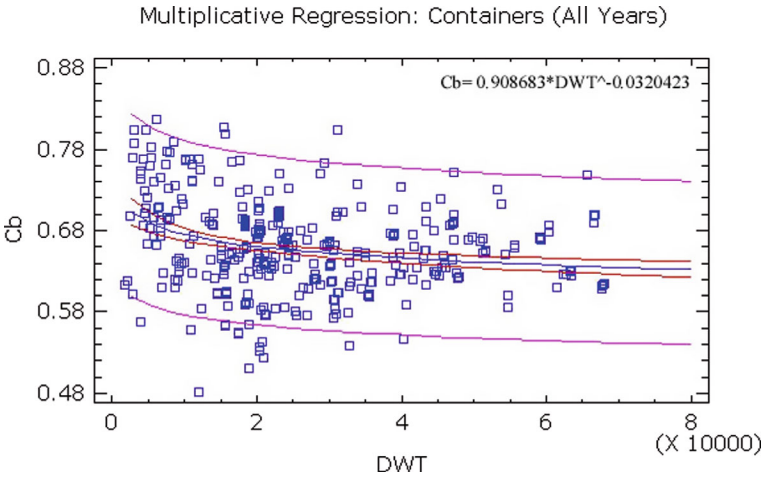


Fig. A.38 Regression analysis of block coefficient C_B versus DWT [tons] for Containerships* (Kalokairinos et al. 2000–2005)

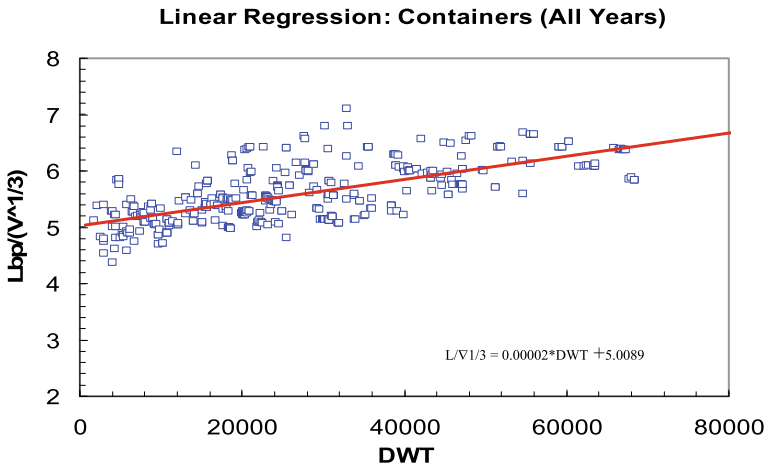


Fig. A.39 Regression analysis of slenderness ratio ($L/\nabla^{1/3}$) versus DWT [tons] for Containerships (Kalokairinos et al. 2000–2005)

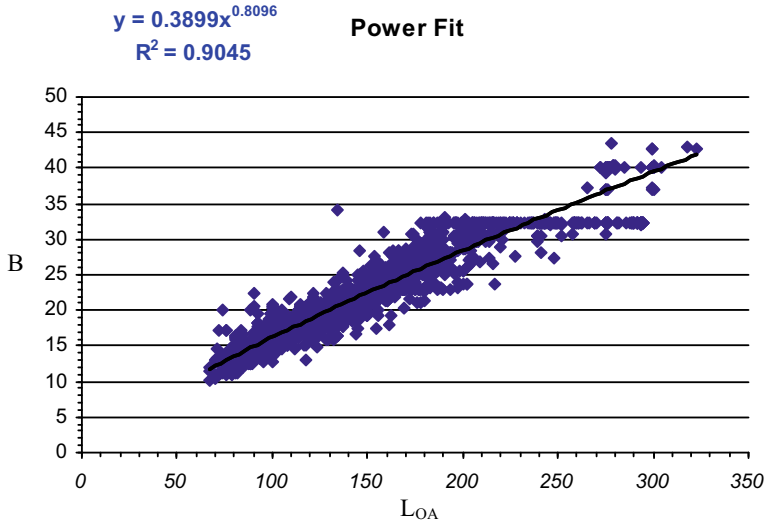


Fig. A.40 Regression analysis of beam B [m] versus L_{OA} [m] for Containerships (Kalokairinos et al. 2000–2005)

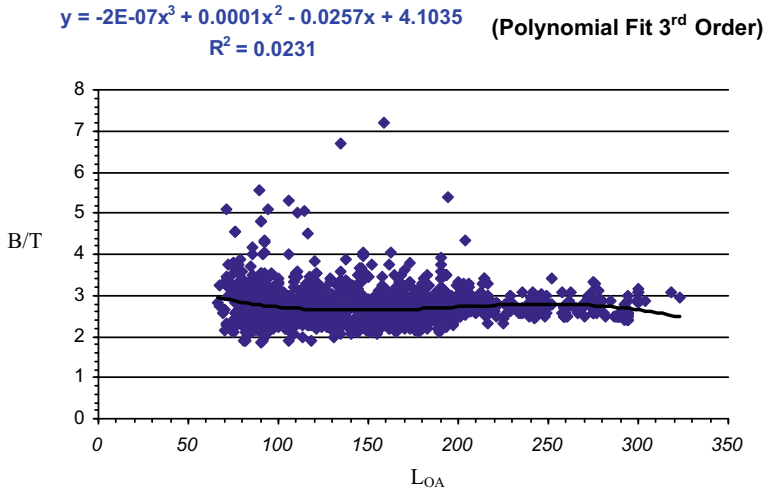


Fig. A.41 Regression analysis of ratio B/T versus L_{OA} [m] for Containerships (Kalokairinos et al. 2000–2005)

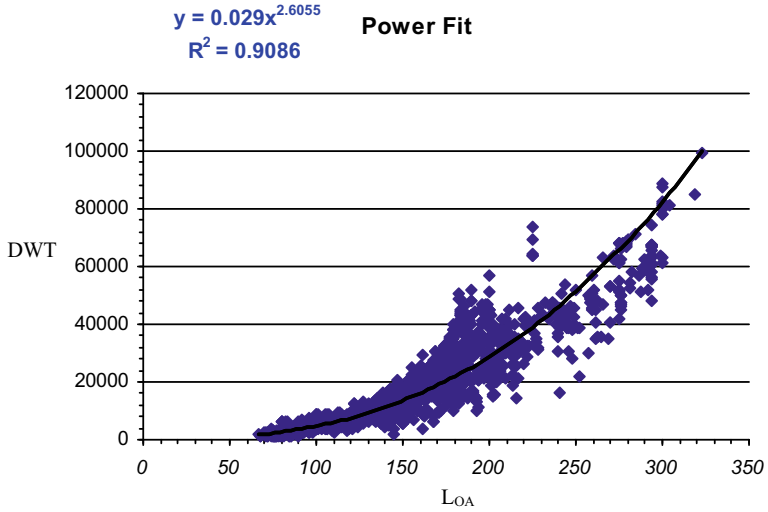


Fig. A.42 Regression analysis of DWT [tons] versus L_{OA} [m] for Containerships (Kalokairinos et al. 2000–2005)

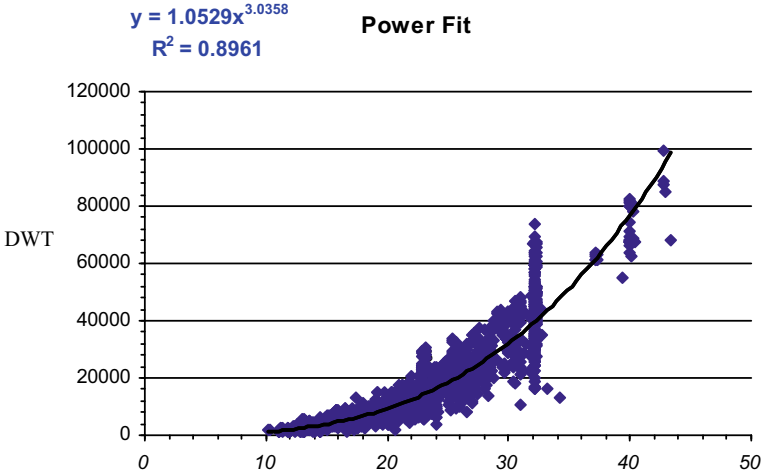


Fig. A.43 Regression analysis of DWT [tons] versus beam B [m] for Containerships (Kalokairinos et al. 2000–2005)

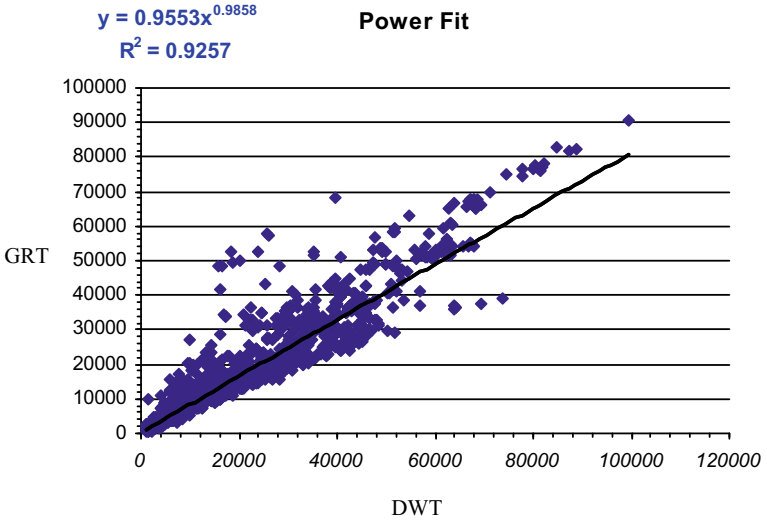


Fig. A.44 Regression analysis of GRT [RT] versus DWT [tons] for Containerships (Kalokairinos et al. 2000–2005)

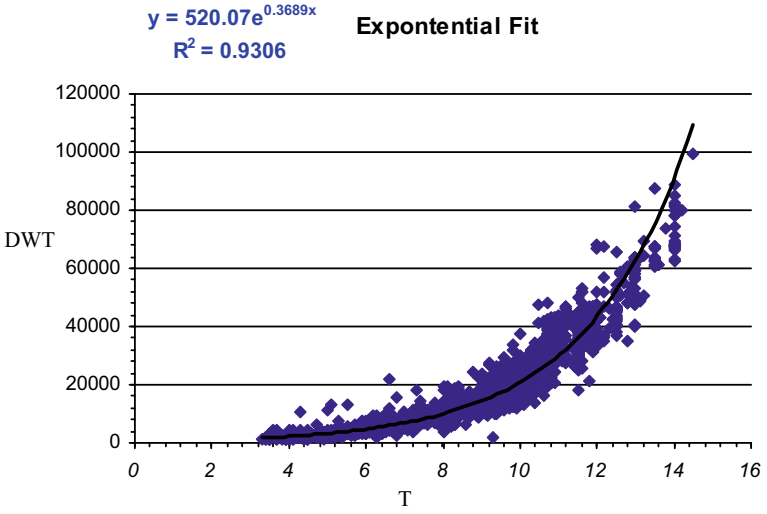


Fig. A.45 Regression analysis of DWT [tons] versus draft T [m] for Containerships (Kalokairinos et al. 2000–2005)

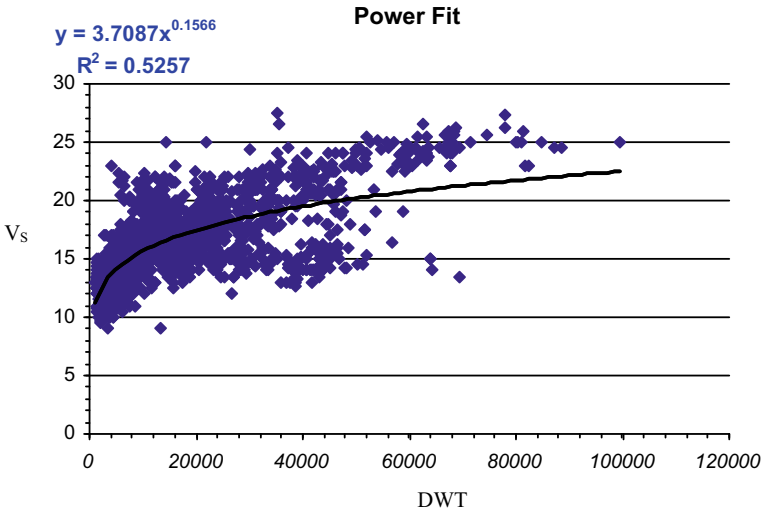


Fig. A.46 Regression analysis of the speed V [knots] versus DWT [tons] for Containerships (Kalokairinos et al. 2000–2005)

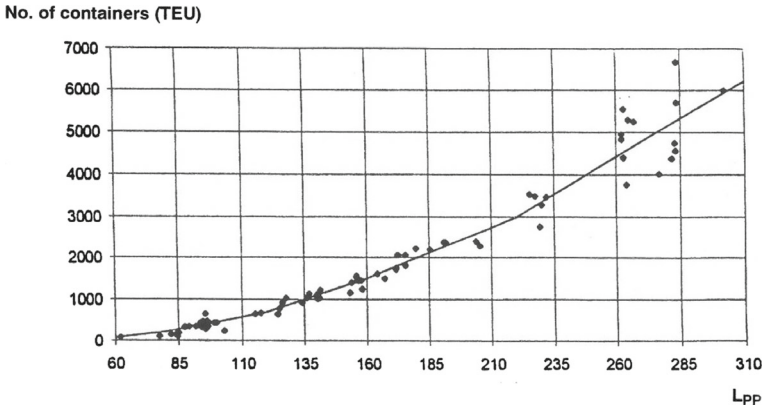


Fig. A.47 Regression analysis of the number of containers versus the length L [m] for containerships according to Kristensen (2000) in Friis et al. (2002)

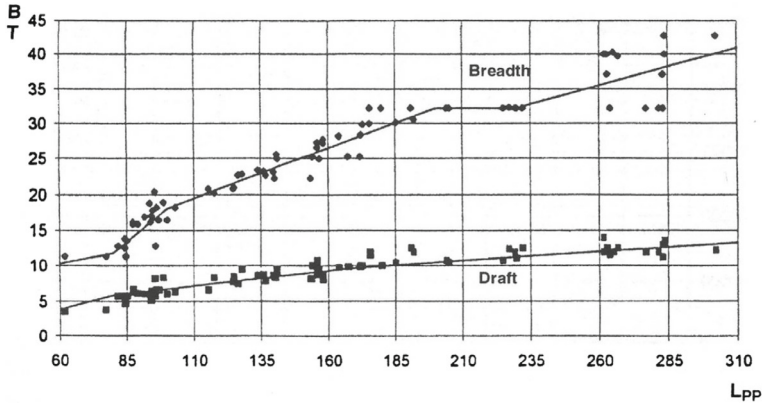


Fig. A.48 Regression analysis of the beam B [m] and the draft T [m] versus the length L [m] for containerships according to Kristensen (2000) in Friis et al. (2002)

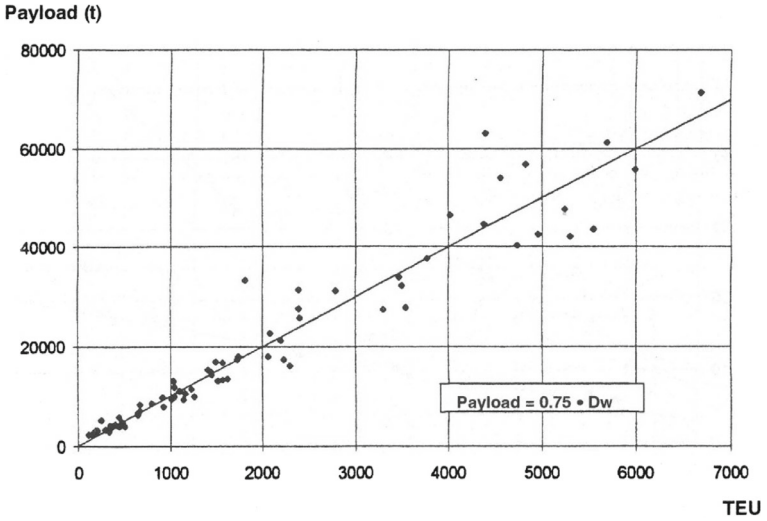


Fig. A.49 Regression analysis of the Payload [tons] versus the number of containers for containerships according to Kristensen (2000) in Friis et al. (2002)

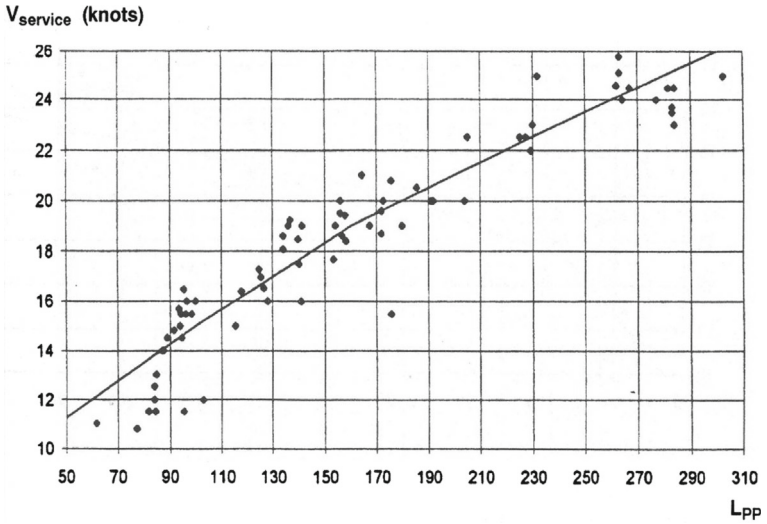


Fig. A.50 Regression analysis of the service speed Vs [knots] versus the length between perpendiculars Lpp [m] for containerships according to Kristensen (2000) in Friis et al. (2002)

A.4 Tankers

See Figs. A.51, A.52, A.53, A.54, A.55, A.56, A.57, A.58, A.59 and A.60.

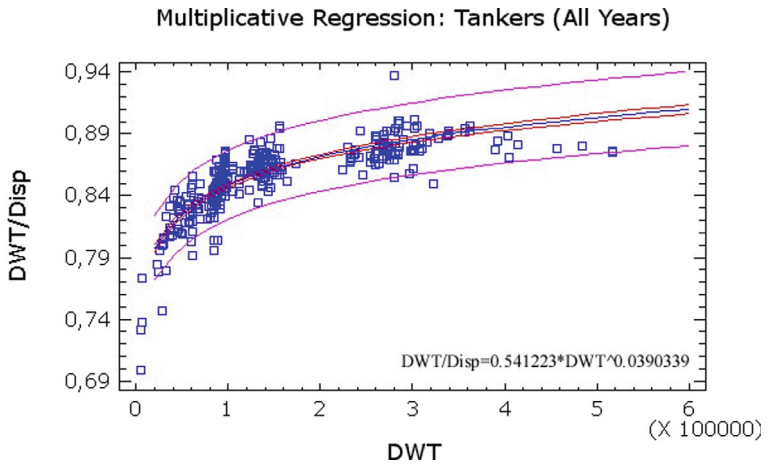


Fig. A.51 Regression analysis of ratio (DWT/Δ) versus DWT [tons] for Tankers* (Kalokairinos et al. 2000–2005)

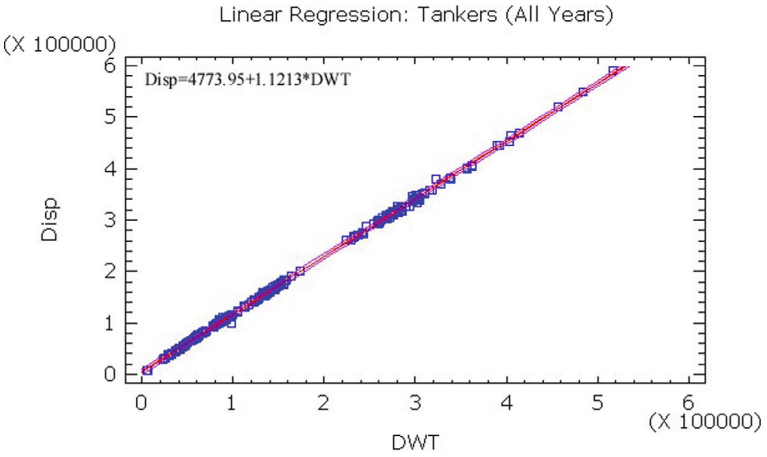


Fig. A.52 Regression analysis of displacement Δ [tons] versus DWT [tons] for Tankers* (Kalokairinos et al. 2000–2005)

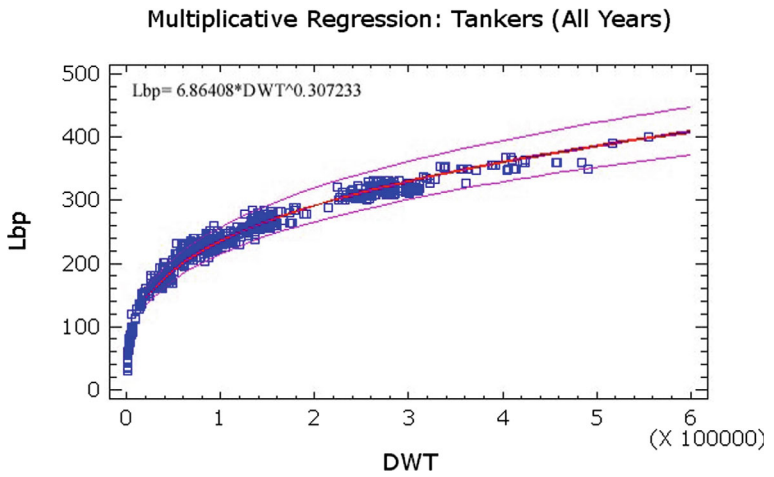


Fig. A.53 Regression analysis of length L_{BP} [m] versus DWT [tons] for Tankers* (Kalokairinos et al. 2000–2005)

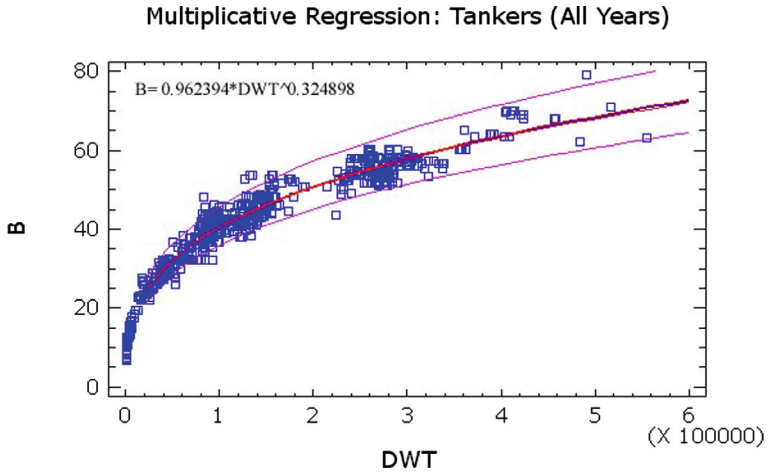


Fig. A.54 Regression analysis of beam B [m] versus DWT [tons] for Tankers* (Kalokairinos et al. 2000–2005)

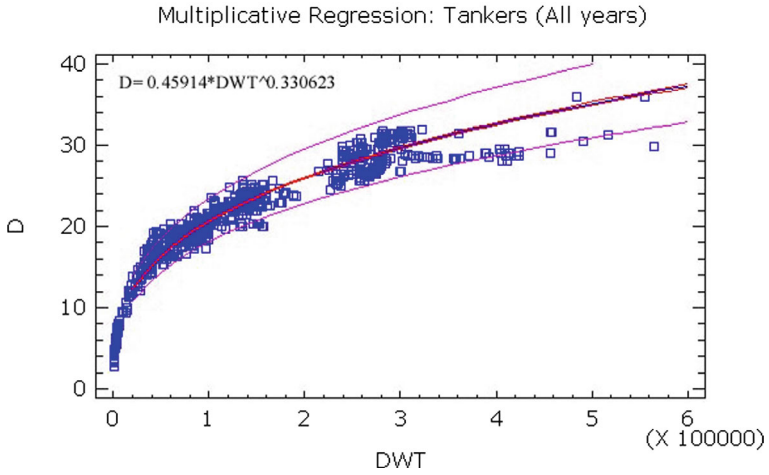


Fig. A.55 Regression analysis of side depth D [m] versus DWT [m] for Tankers* (Kalokairinos et al. 2000–2005)

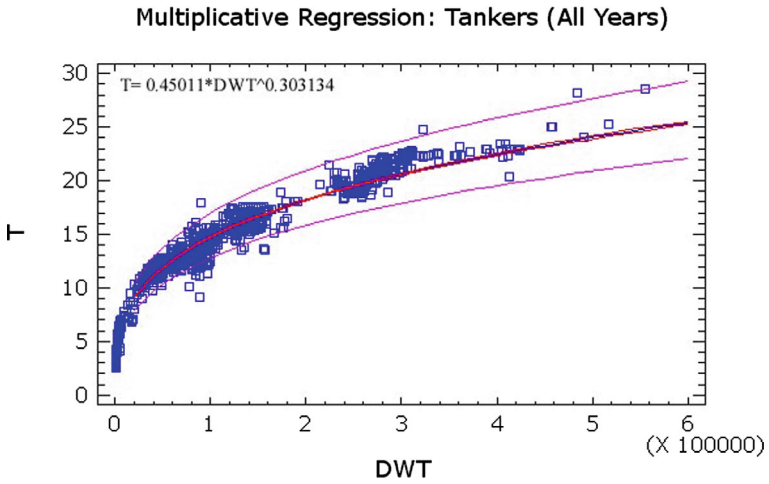


Fig. A.56 Regression analysis of draft T [m] versus DWT [tons] for Tankers* (Kalokairinos et al. 2000–2005)

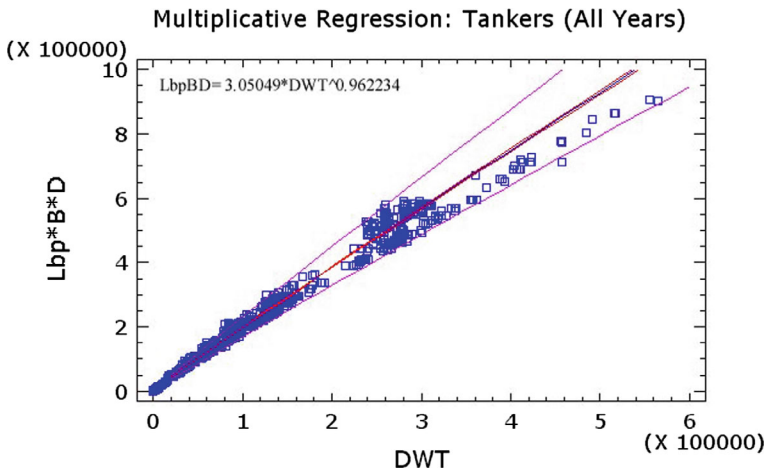


Fig. A.57 Regression analysis of volumetric product (L × B × D) [m^3] versus DWT [tons] for Tankers* (Kalokairinos et al. 2000–2005)

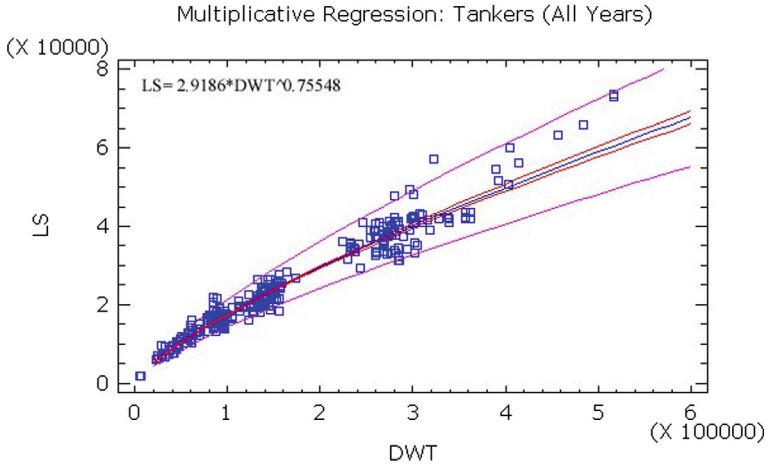


Fig. A.58 Regression analysis of lightship weight (LS) [tons] versus DWT [tons] for Tankers* (Kalokairinos et al. 2000–2005)

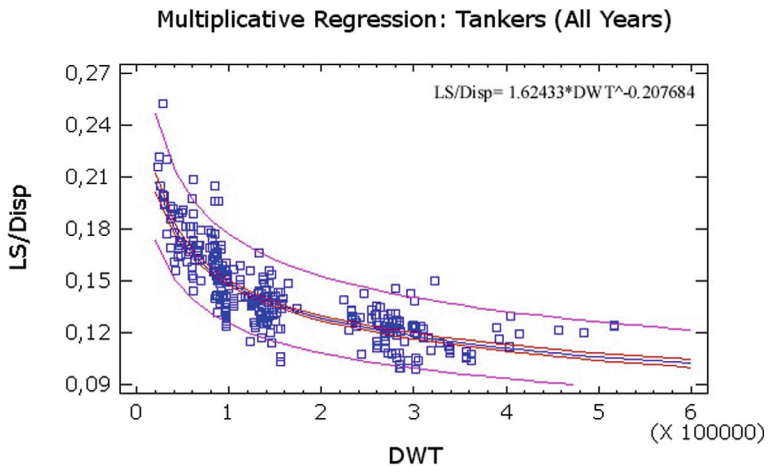


Fig. A.59 Regression analysis of ratio (LS/Δ) versus DWT [tons] for Tankers* (Kalokairinos et al. 2000–2005)

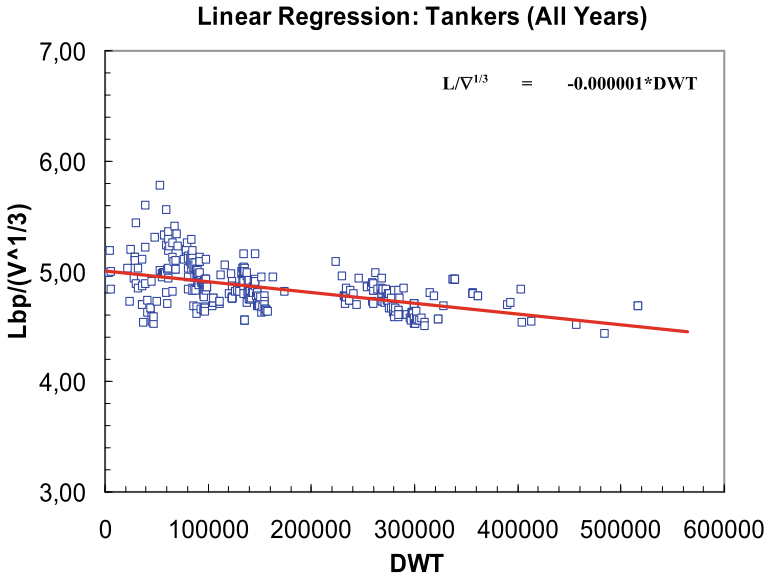


Fig. A.60 Regression analysis of slenderness ratio ($L/\nabla^{1/3}$) versus DWT [tons] for Tankers (Kalokairinos et al. 2000–2005)

A.5 Product Carriers

See Figs. A.61, A.62, A.63, A.64, A.65, A.66, A.67, A.68, A.69, A.70, A.71, A.72 and A.73.

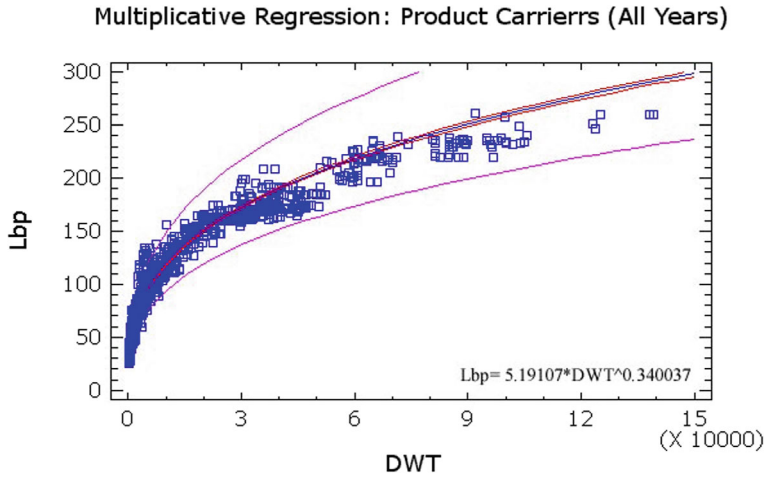


Fig. A.61 Regression analysis of length L_{BP} [m] versus DWT [tons] for Product Carriers* (Kalokairinos et al. 2000–2005)

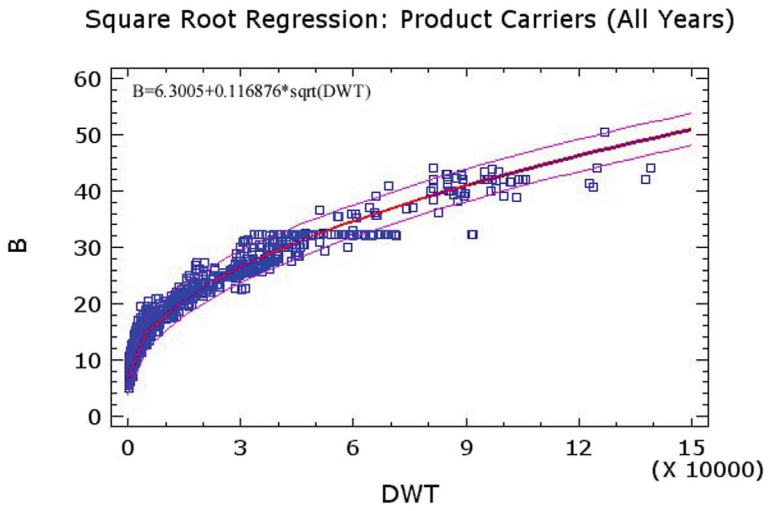


Fig. A.62 Regression analysis of beam B [m] versus DWT [m] for Product Carriers* (Kalokairinos et al. 2000–2005)

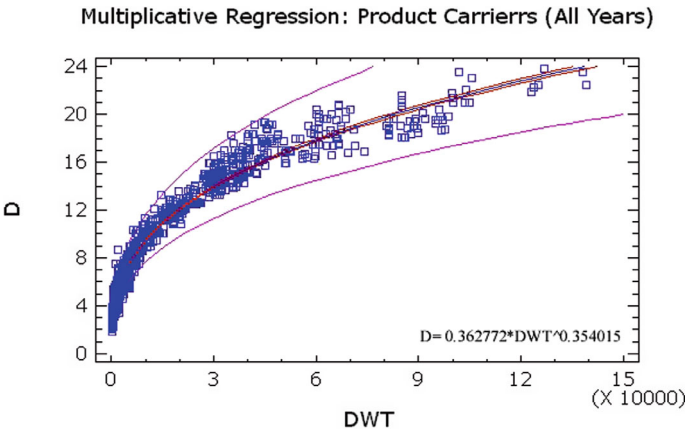


Fig. A.63 Regression analysis of side depth D [m] versus DWT [m] for Product Carriers* (Kalokairinos et al. 2000–2005)

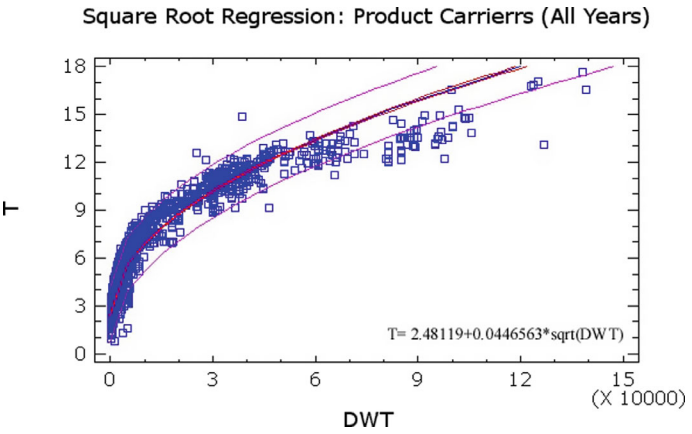


Fig. A.64 Regression analysis of draft T [m] versus DWT [tons] for Product Carriers* (Kalokairinos et al. 2000–2005)

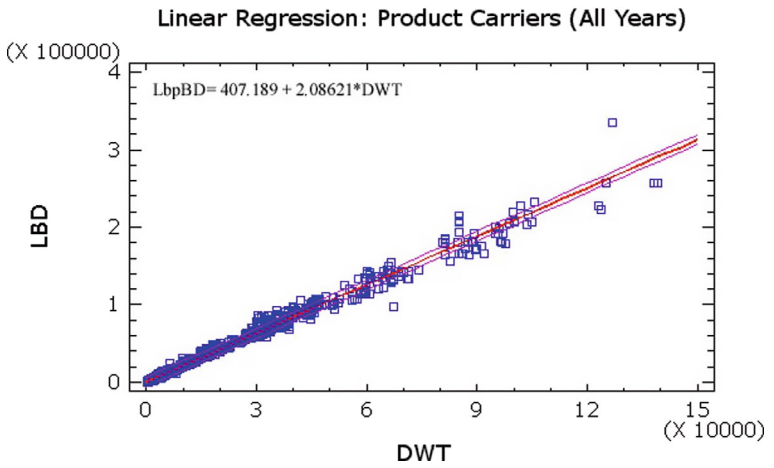


Fig. A.65 Regression analysis of volumetric product (L × B × D) [m^3] versus DWT [tons] for Product Carriers* (Kalokairinos et al. 2000–2005)

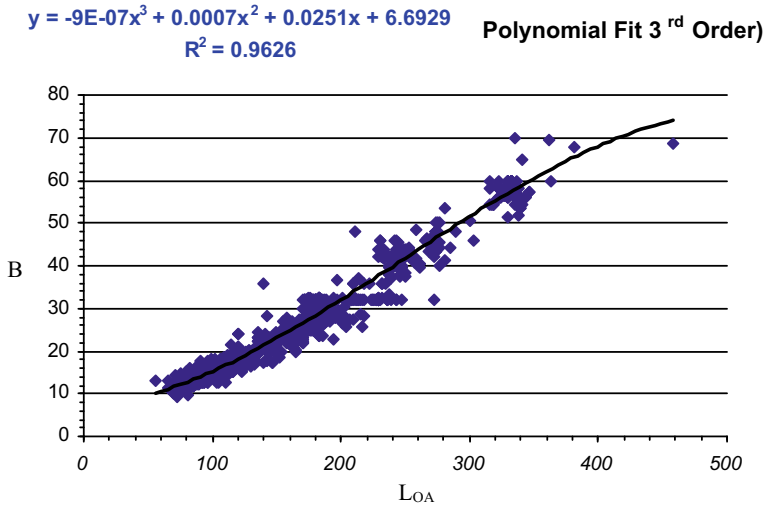


Fig. A.66 Regression analysis of beam B [m] versus length L_{OA} [m] for Product Carriers (Kalokairinos et al. 2000–2005)

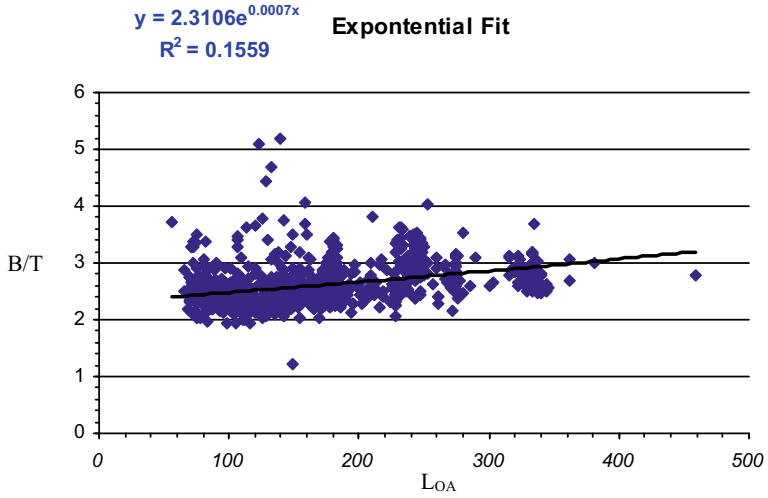


Fig. A.67 Regression analysis of ratio B/T versus the length L_{OA} [m] for Product Carriers (Kalokairinos et al. 2000–2005)

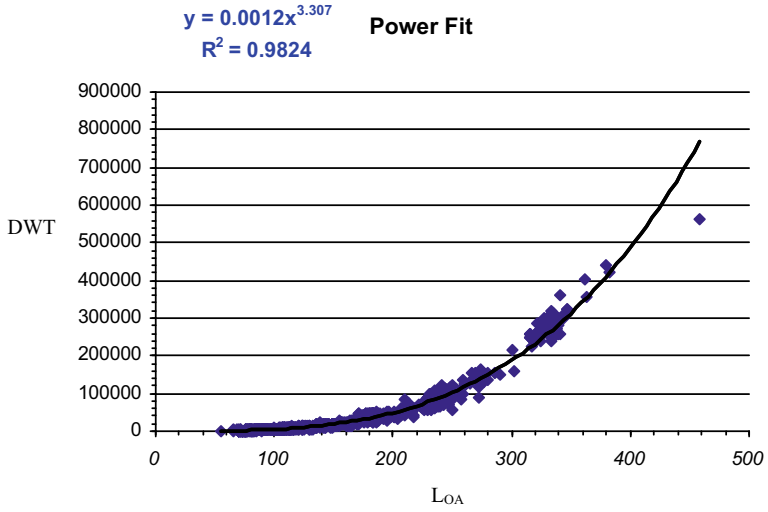


Fig. A.68 Regression analysis of DWT [tons] versus the length L_{OA} [m] for Product Carriers (Kalokairinos et al. 2000–2005)

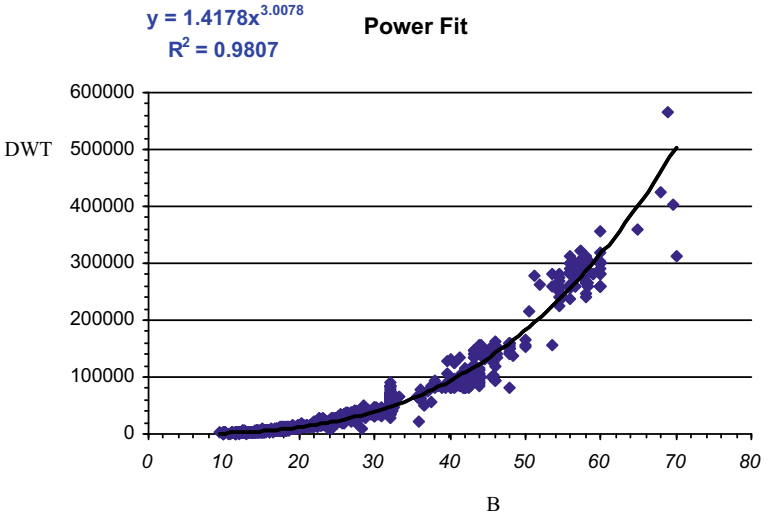


Fig. A.69 Regression analysis of DWT [tons] versus beam B [m] for Product Carriers (Kalokairinos et al. 2000–2005)

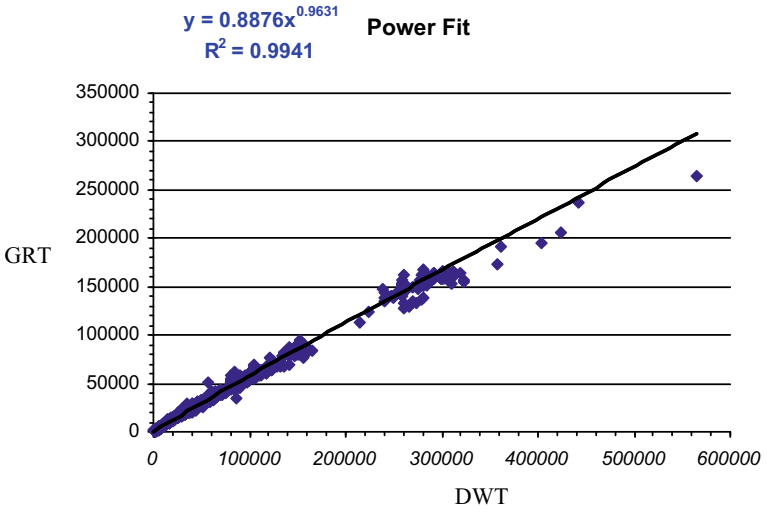


Fig. A.70 Regression analysis of GRT [RT] versus DWT [tons] for Product Carriers (Kalokairinos et al. 2000–2005)

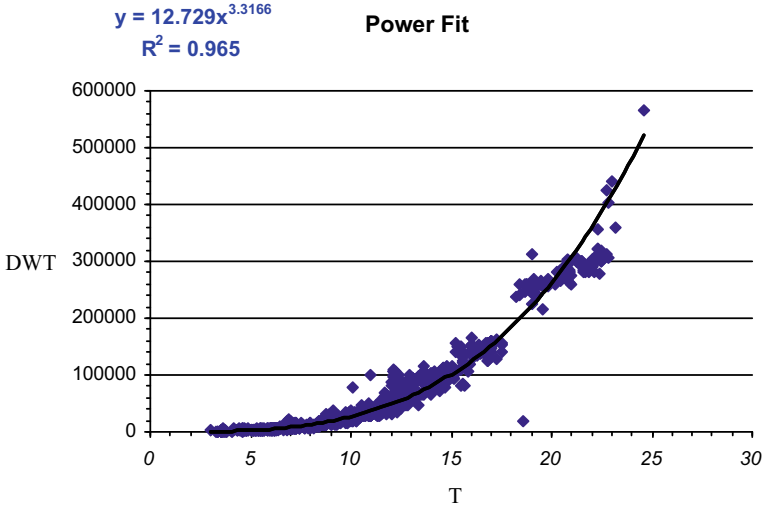


Fig. A.71 Regression analysis of DWT tons versus draft T [m] for Product Carriers (Kalokairinos et al. 2000–2005)

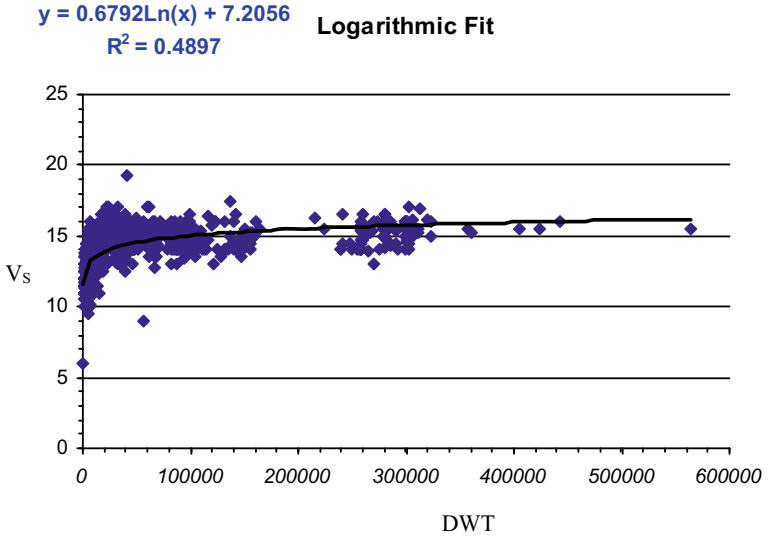


Fig. A.72 Regression analysis of the speed V [knots] versus DWT [tons] for Product Carriers (Kalokairinos et al. 2000–2005)

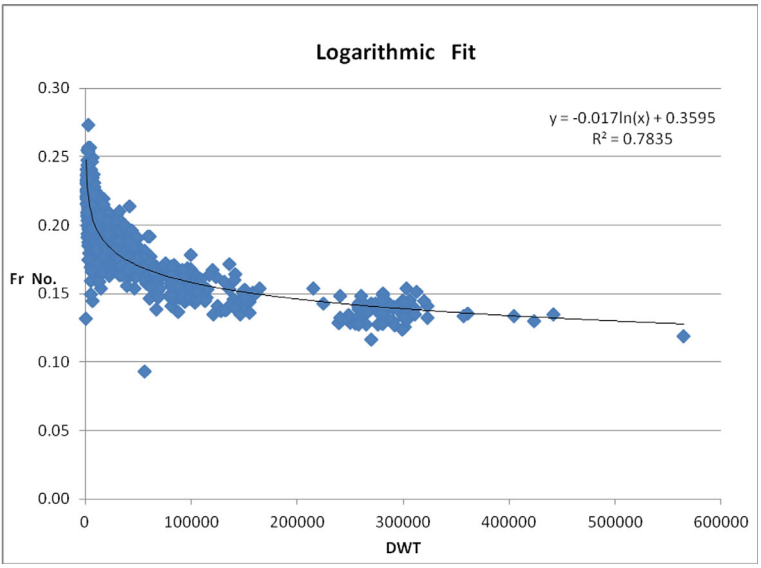


Fig. A.73 Regression analysis of the Froude no. versus DWT [tons] for Product Carriers (Kalokairinos et al. 2000–2005)

A.6 Chemical Carriers

See Figs. A.74, A.75, A.76, A.77, A.78, A.79, A.80 and A.81.

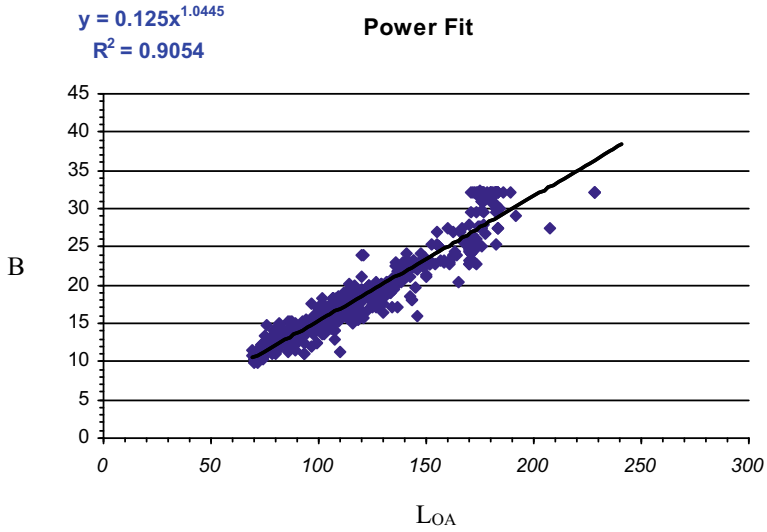


Fig. A.74 Regression analysis of beam B [m] versus length L_{OA} [m] for Chemical Carriers (Kalokairinos et al. 2000–2005)

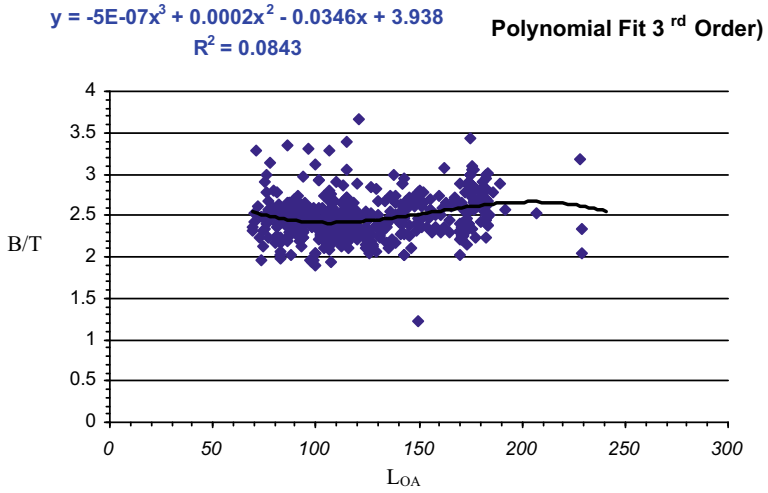


Fig. A.75 Regression analysis of ratio B/T versus length L_{OA} [m] for Chemical Carriers (Kalokairinos et al. 2000–2005)

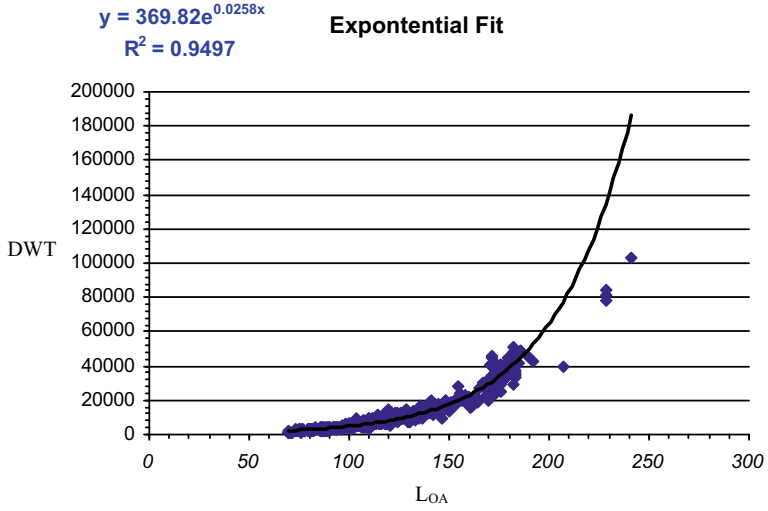


Fig. A.76 Regression analysis of DWT [tons] versus length L_{OA} [m] for Chemical Carriers (Kalokairinos et al. 2000–2005)

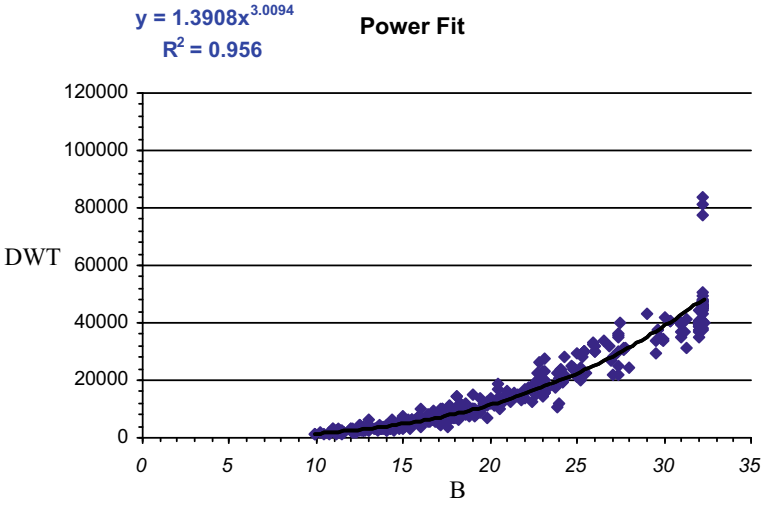


Fig. A.77 Regression analysis of DWT [tons] versus beam B [m] for Chemical Carriers (Kalokairinos et al. 2000–2005)

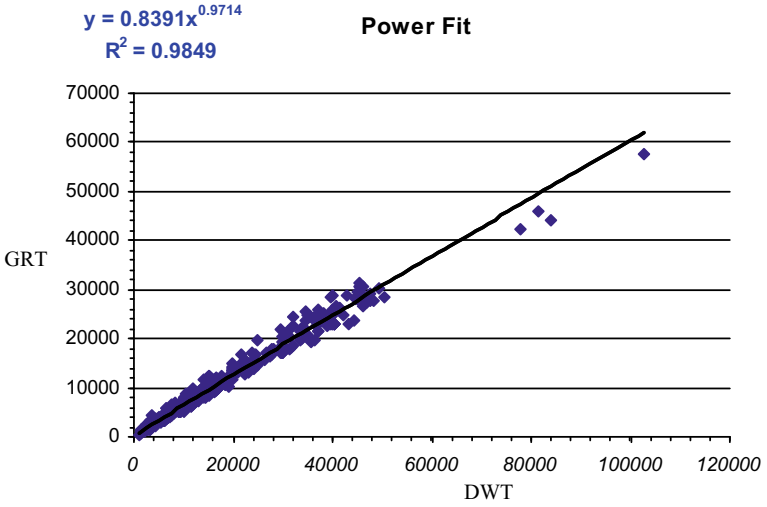


Fig. A.78 Regression analysis of GRT [RT] versus DWT [tons] for Chemical Carriers (Kalokairinos et al. 2000–2005)

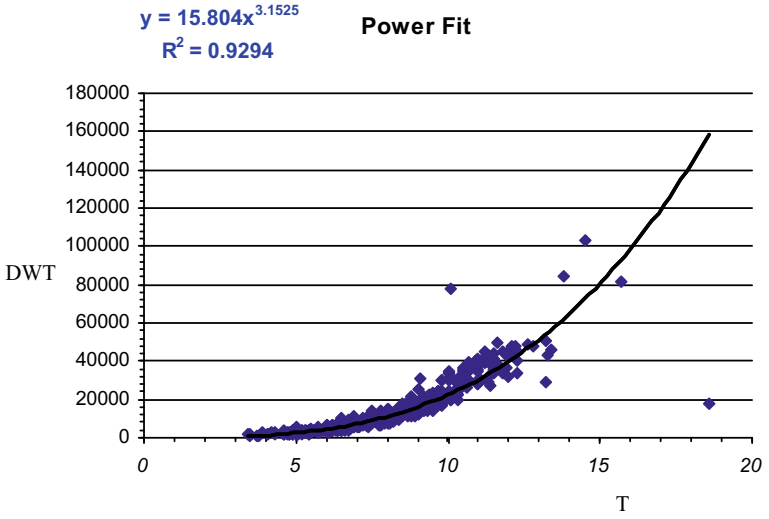


Fig. A.79 Regression analysis of DWT [tons] versus draft T [m] for Chemical Carriers (Kalokairinos et al. 2000–2005)

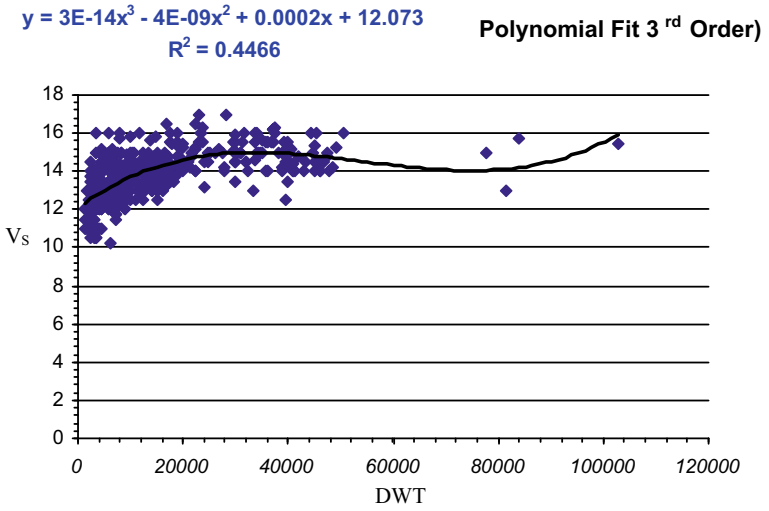


Fig. A.80 Regression analysis of the speed V [knots] versus DWT [tons] for Chemical Carriers (Kalokairinos et al. 2000–2005)

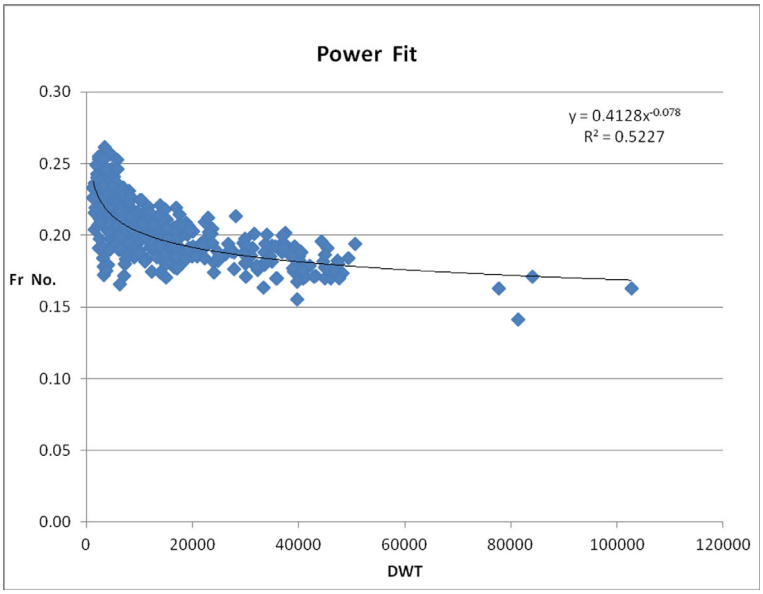


Fig. A.81 Regression analysis of the Froude number versus DWT [tons] for Chemical Carriers (Kalokairinos et al. 2000–2005)

A.7 General Cargo Carriers

See Figs. A.82, A.83, A.84, A.85, A.86, A.87 and A.88.

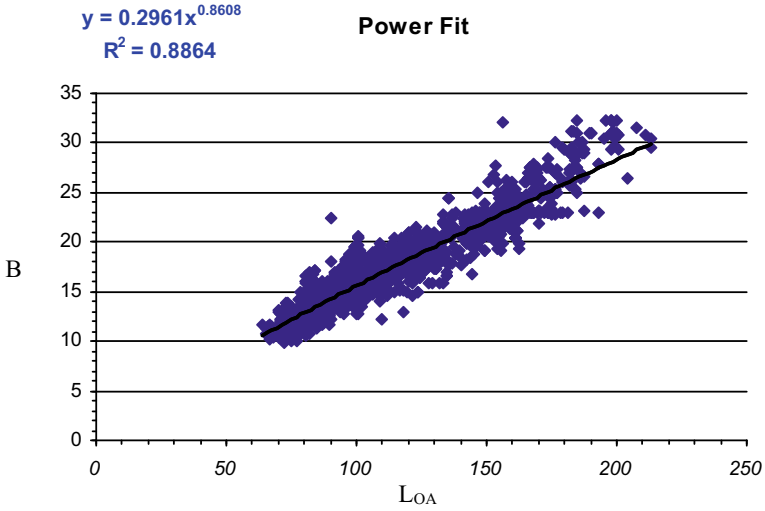


Fig. A.82 Regression analysis of beam B [m] versus length L_{OA} [m] for General Cargo Carriers (Kalokairinos et al. 2000–2005)

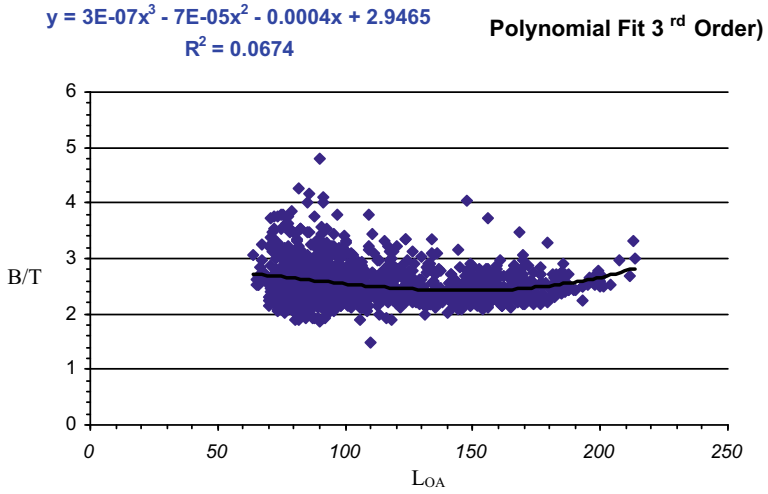


Fig. A.83 Regression analysis of ratio B/T versus length L_{OA} [m] for General Cargo Carriers (Kalokairinos et al. 2000–2005)

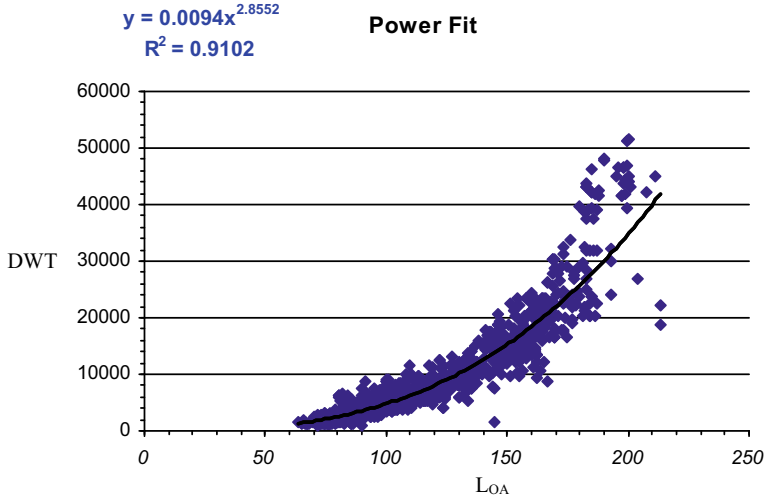


Fig. A.84 Regression analysis of the DWT [tons] versus length L_{OA} [m] for General Cargo Carriers (Kalokairinos et al. 2000–2005)

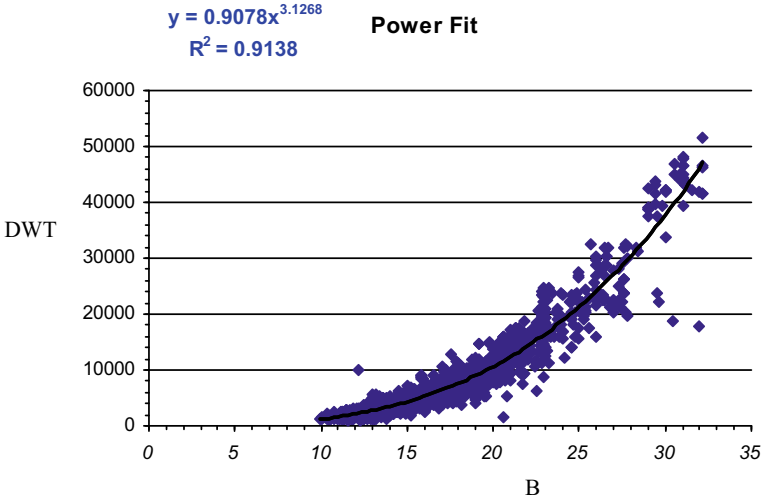


Fig. A.85 Regression analysis of DWT [tons] versus the beam B [m] for General Cargo Carriers (Kalokairinos et al. 2000–2005)

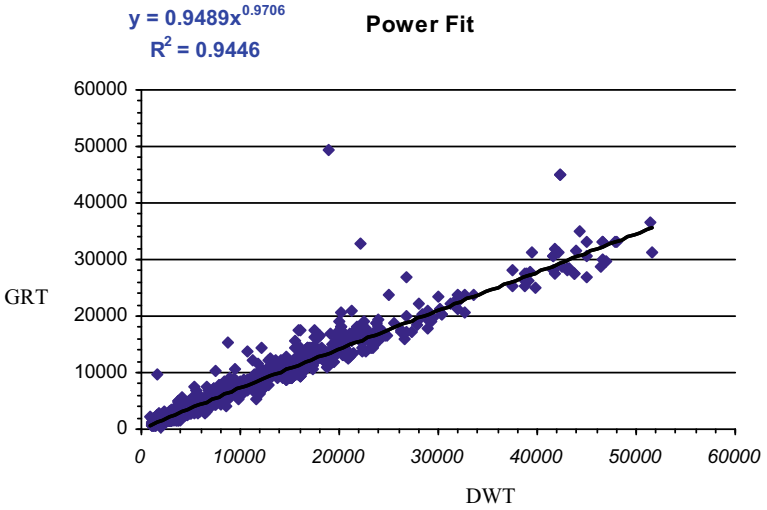


Fig. A.86 Regression analysis of GRT [RT] versus DWT [tons] for General Cargo Carriers (Kalokairinos et al. 2000–2005)

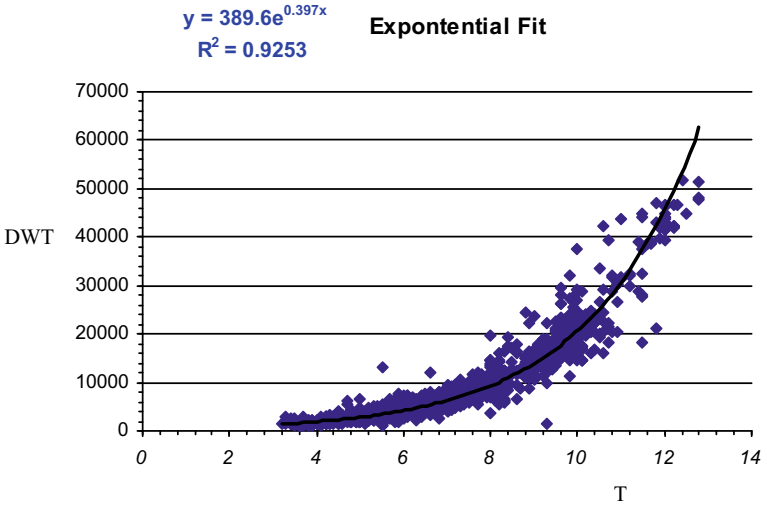


Fig. A.87 Regression analysis of DWT [tons] versus the draft T [m] for General Cargo Carriers (Kalokairinos et al. 2000–2005)

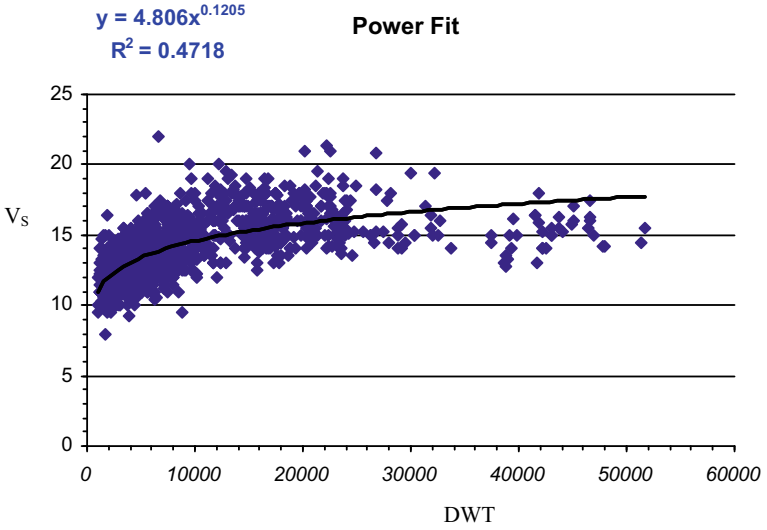


Fig. A.88 Regression analysis of the speed V [knots] versus DWT [tons] for General Cargo Carriers (Kalokairinos et al. 2000–2005)

A.8 Ro-Ro Cargo Ships

See Figs. A.89, A.90, A.91, A.92, A.93, A.94, A.95, A.96, A.97 and A.98.

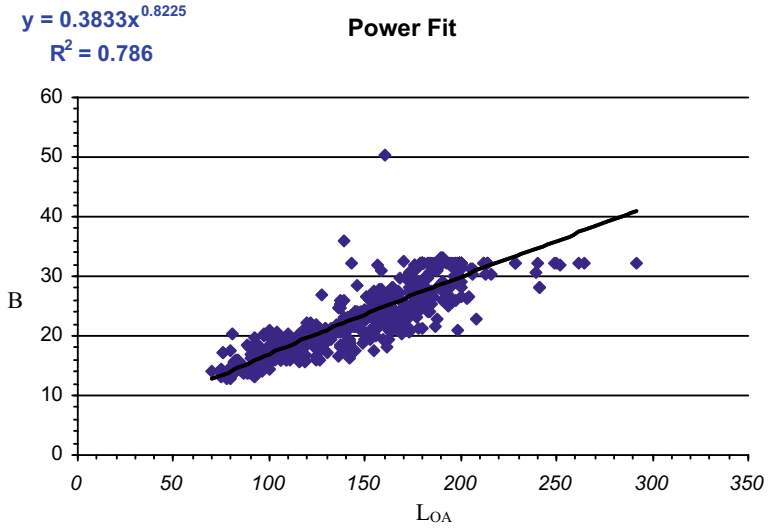


Fig. A.89 Regression analysis of beam B [m] versus length L_{OA} [m] for Ro-Ro Cargo Ships (Kalokairinos et al. 2000–2005)

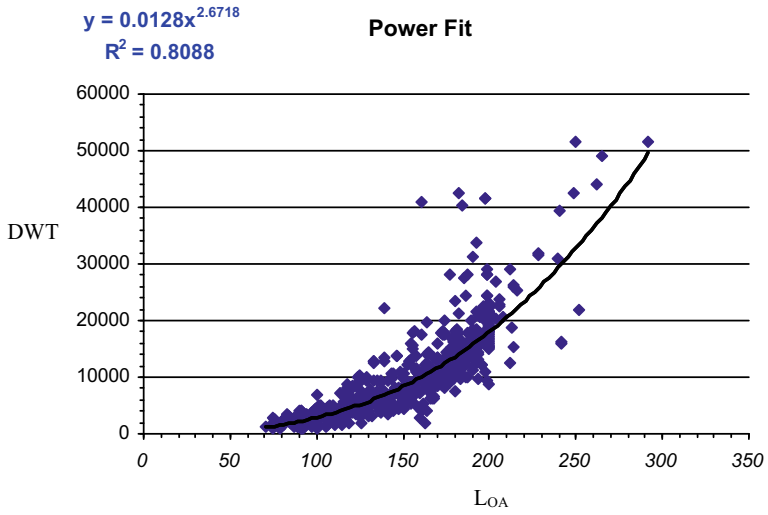


Fig. A.90 Regression analysis of DWT [tons] versus length L_{OA} [m] for Ro-Ro Cargo Ships (Kalokairinos et al. 2000–2005)

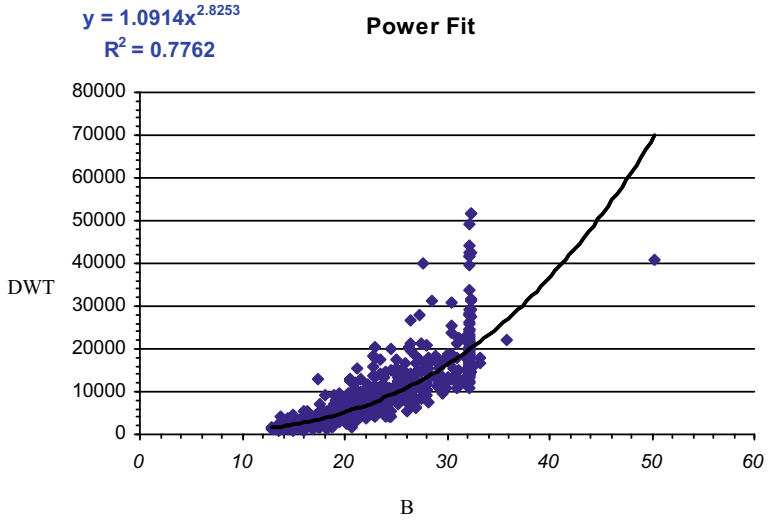


Fig. A.91 Regression analysis of DWT [tons] versus beam B [m] for Ro-Ro Cargo Ships (Kalokairinos et al. 2000–2005)

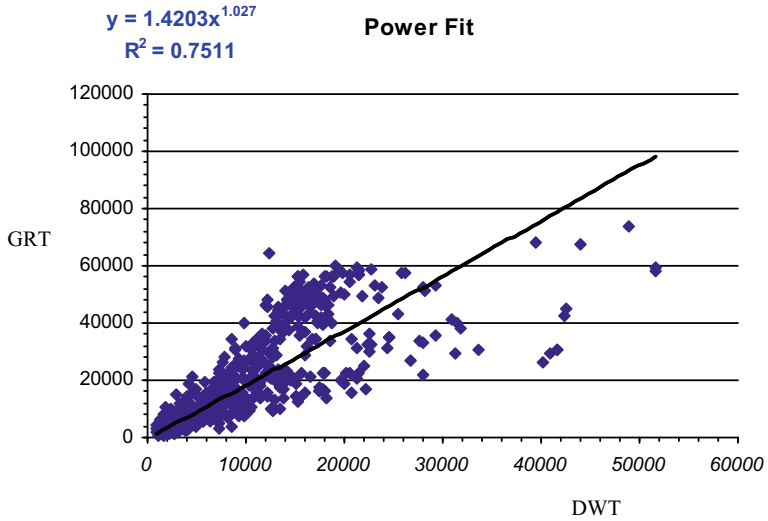


Fig. A.92 Regression analysis of GRT [RT] versus DWT [tons] for Ro-Ro Cargo Ships (Kalokairinos et al. 2000–2005)

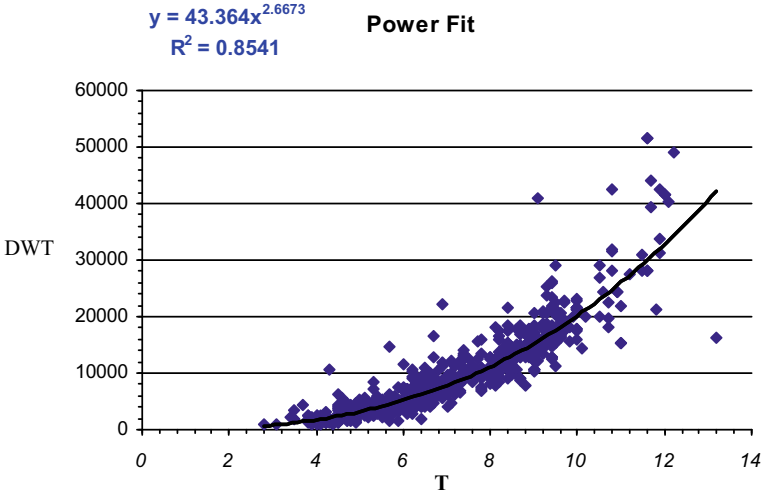


Fig. A.93 Regression analysis of DWT [tons] versus draft T [m] for Ro-Ro Cargo Ships (Kalokairinos et al. 2000–2005)

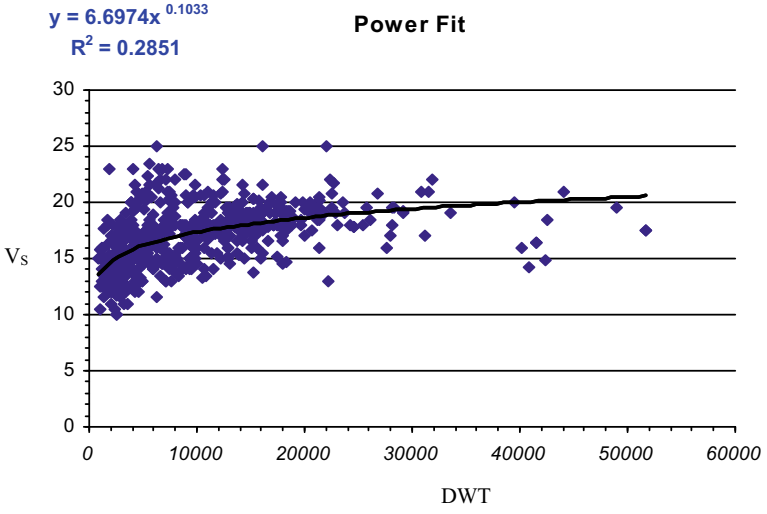


Fig. A.94 Regression analysis of the speed V [knots] versus DWT [tons] for Ro-Ro Cargo Ships (Kalokairinos et al. 2000–2005)

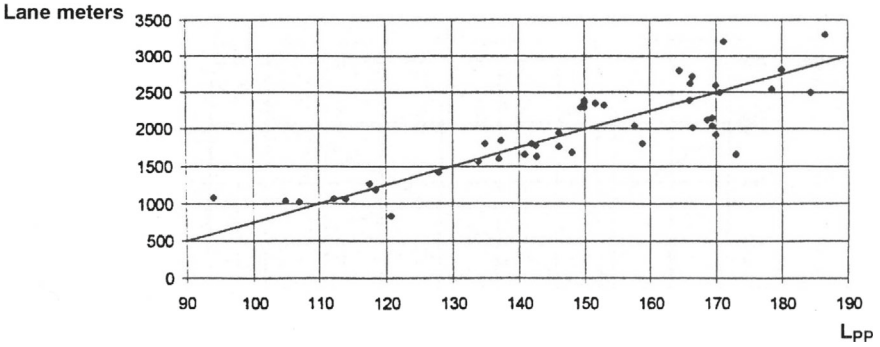


Fig. A.95 Regression analysis of vehicles' lane length [m] versus the length L [m] for Ro-Ro cargo ships according to Kristensen (2000) in Friis et al. (2002)

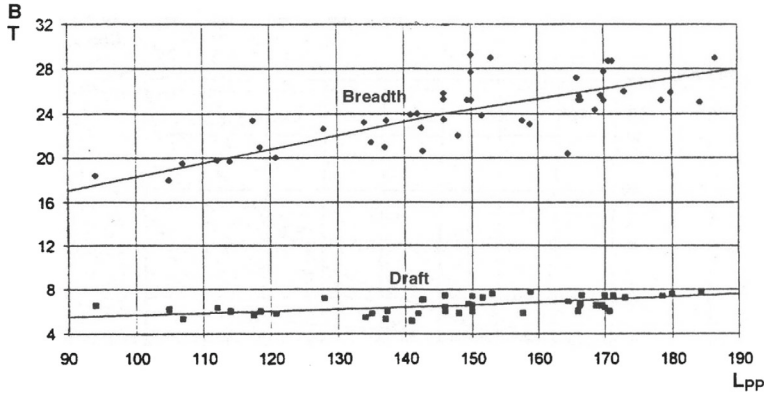


Fig. A.96 Regression analysis of the beam B [m] and the draft T [m] versus the length L [m] for Ro-Ro cargo ships according to Kristensen (2000) in Friis et al. (2002)

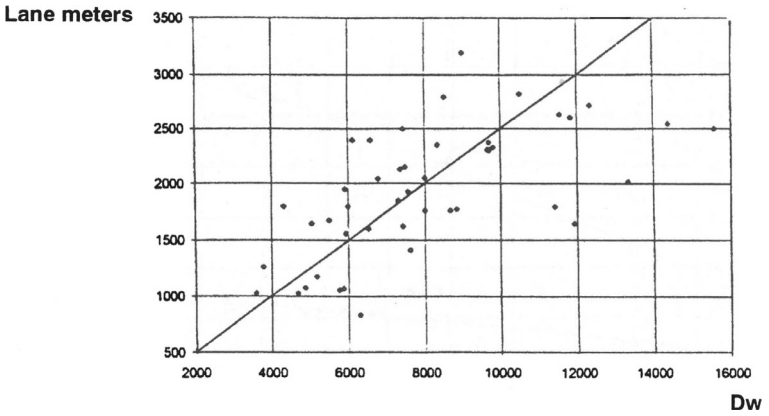


Fig. A.97 Regression analysis of the vehicles' lanes length [m] versus the DWT [tons] for Ro-Ro cargo ships according to Kristensen (2000) in Friis et al. (2002)

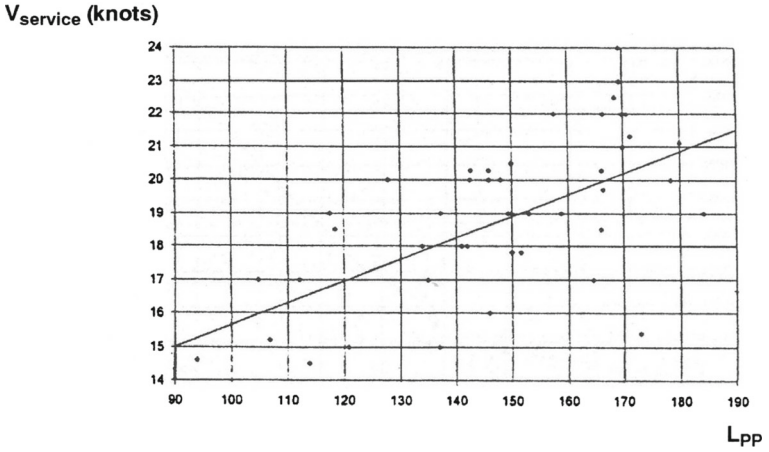


Fig. A.98 Regression analysis of the service speed Vs [knots] versus the ship length L [m] for Ro-Ro cargo ships according to Kristensen (2000) in Friis et al. (2002)

A.9 Ro-Ro Passenger Ferries

See Figs. A.99, A.100, A.101, A.102, A.103, A.104 and A.105.

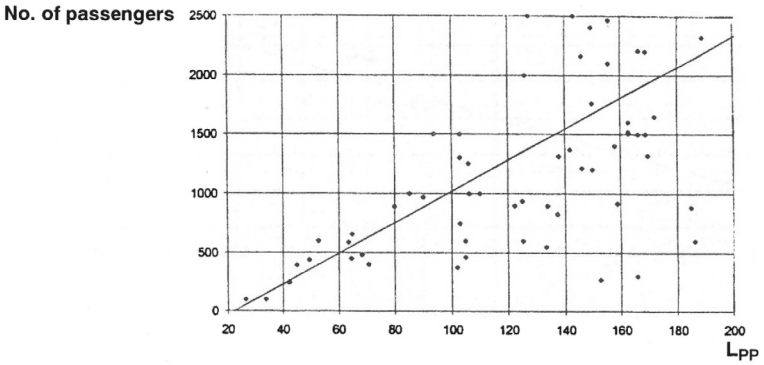


Fig. A.99 Regression analysis of the number of passengers versus the ship length L [m] for RoPAX ships according to Kristensen (2000) in Friis et al. (2002)

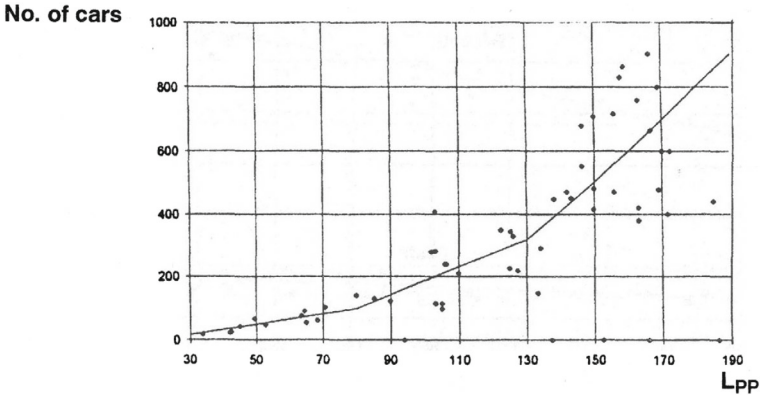


Fig. A.100 Regression analysis of the number of vehicles versus the ship length L [m] for RoPAX ships according to Kristensen (2000) in Friis et al. (2002)

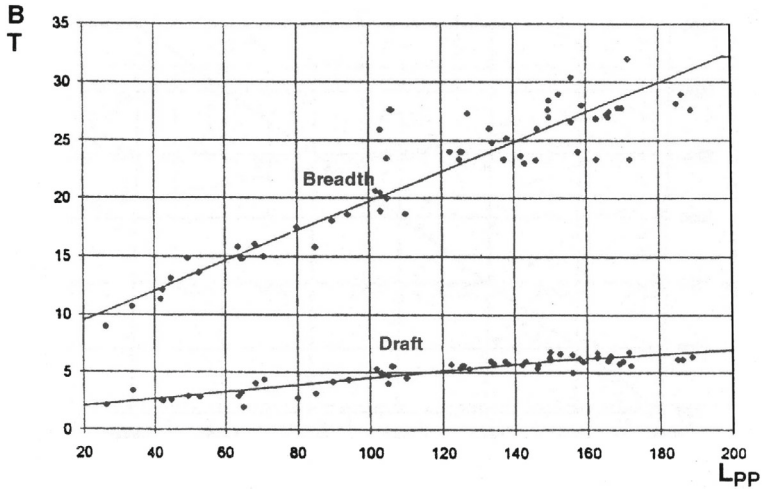


Fig. A.101 Regression analysis of the beam B [m] and the draft T [m] versus the length L [m] for RoPAX ships according to Kristensen (2000) in Friis et al. (2002)

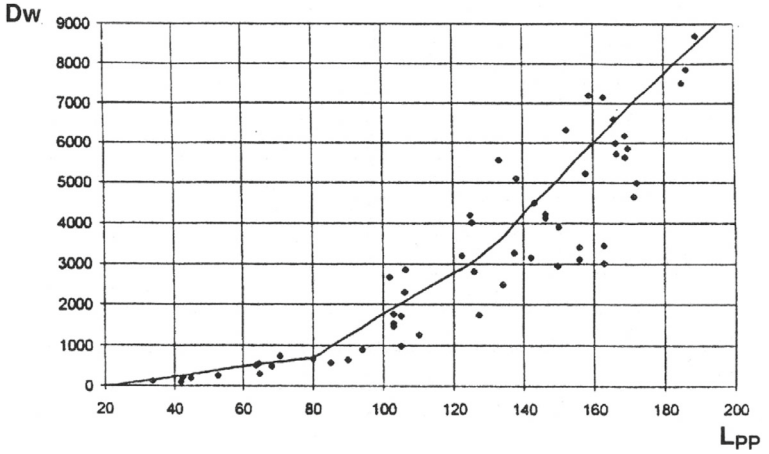


Fig. A.102 Regression analysis of the DWT [tons] versus the ship length L [m] for RoPAX ships according to Kristensen (2000) in Friis et al. (2002)

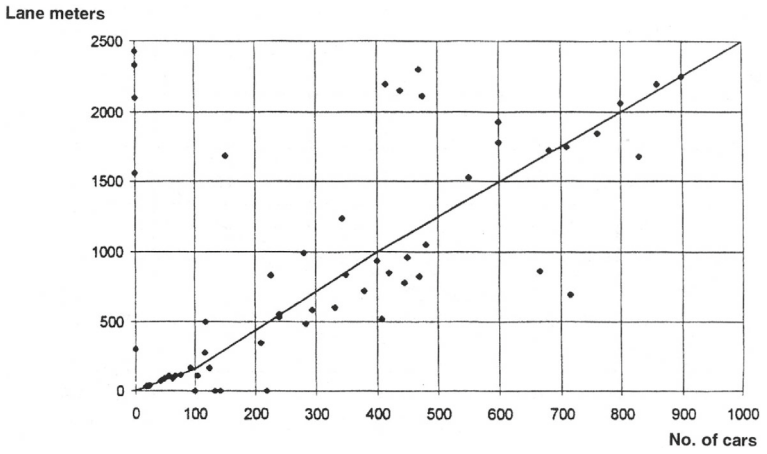


Fig. A.103 Regression analysis of the vehicles' lane length [m] versus the number of vehicles for RoPAX ships according to Kristensen (2000) in Friis et al. (2002)

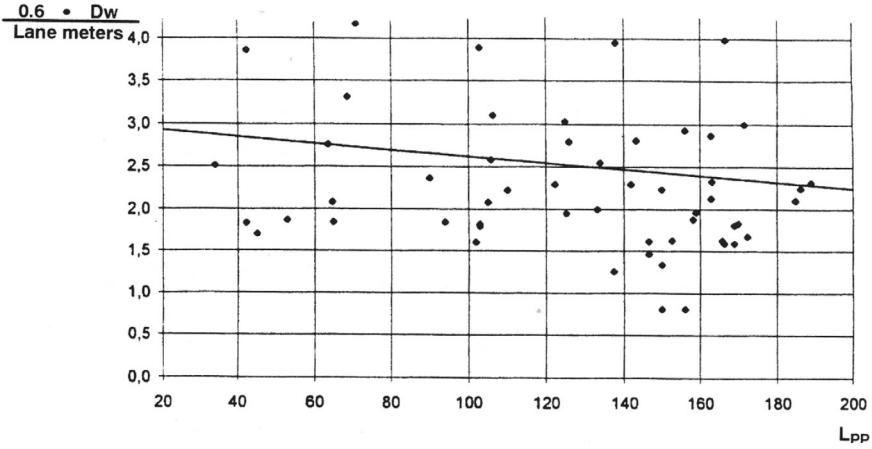


Fig. A.104 Regression analysis of the ratio of 60% of DWT [tons] to the vehicle lane length [m] versus the length L [m] for RoPAX ships according to Kristensen (2000) in Friis et al. (2002)

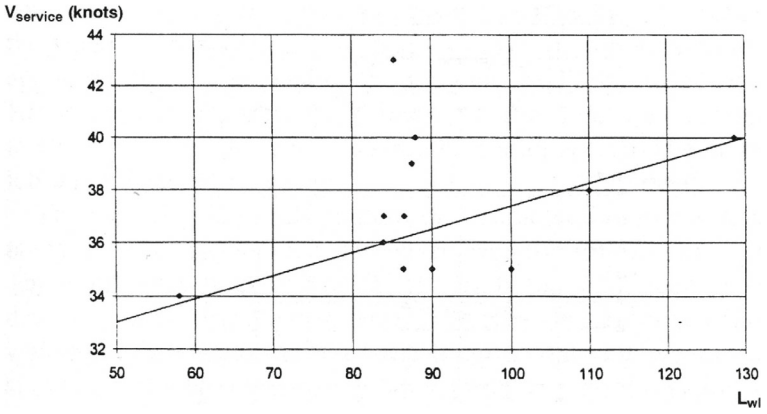


Fig. A.105 Regression analysis of the service speed Vs [knots] versus the length L [m] for RoPAX ships according to Kristensen (2000) in Friis et al. (2002)

A.10 Single-Hull Fast Ferries

See Figs. A.106, A.107, A.108, A.109, A.110 and A.111.

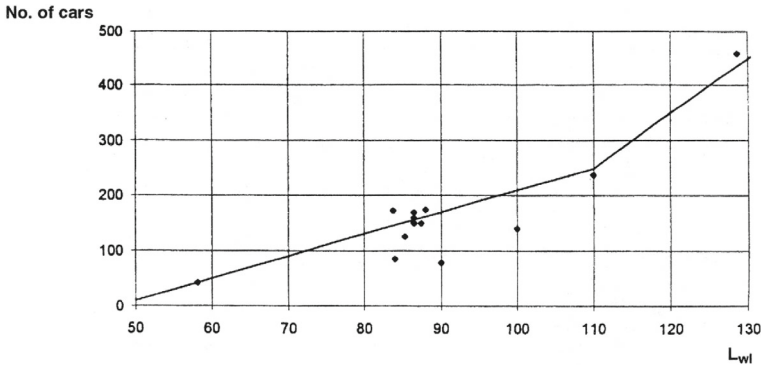


Fig. A.106 Regression analysis of the vehicle number versus the length L_{wl} [m] for fast single-hull ferries according to Kristensen (2000) in Friis et al. (2002)

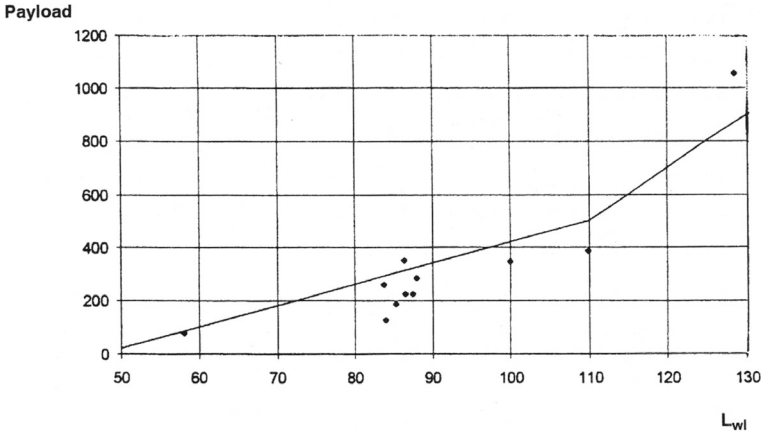


Fig. A.107 Regression analysis of the Payload [tons] versus the length L_{wl} [m] for fast single-hull ferries according to Kristensen (2000) in Friis et al. (2002)

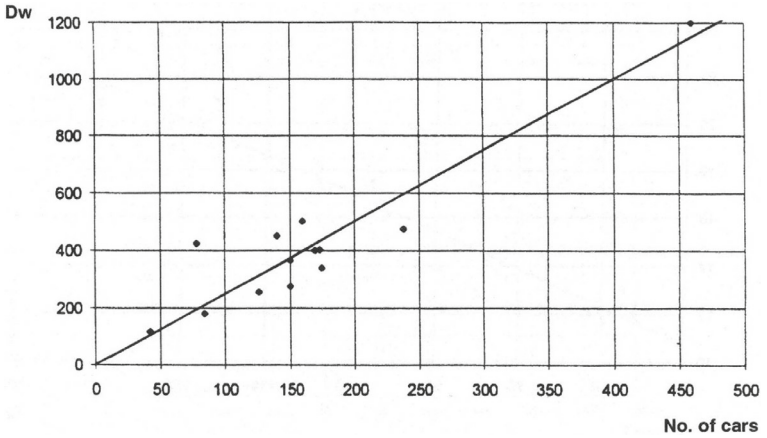


Fig. A.108 Regression analysis of the DWT [tons] versus the vehicle number for fast single-hull ferries according to Kristensen (2000) in Friis et al. (2002)

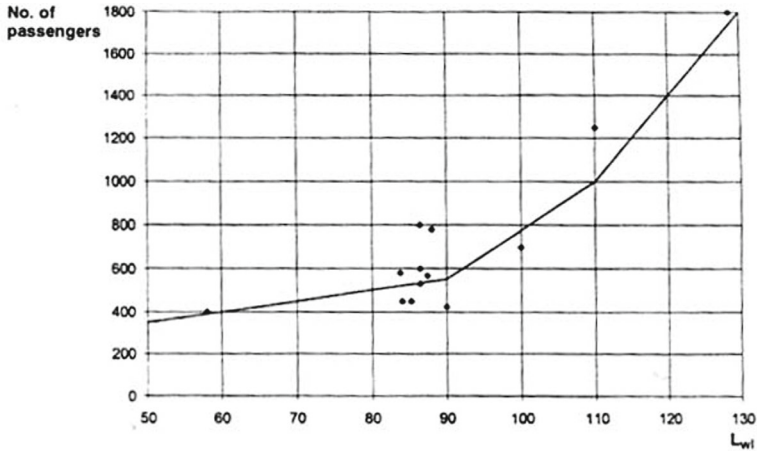


Fig. A.109 Regression analysis of the number of passengers versus the length L_{wl} [m] for fast single-hull ferries according to Kristensen (2000) in Friis et al. (2002)

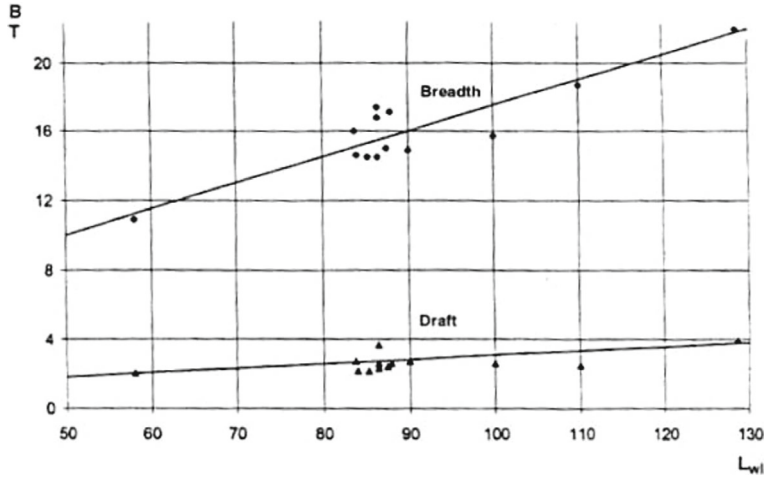


Fig. A.110 Regression analysis of the beam B [m] and draft T [m] versus the length L_{wl} [m] for fast single-hull ferries according to Kristensen (2000) in Friis et al. (2002)

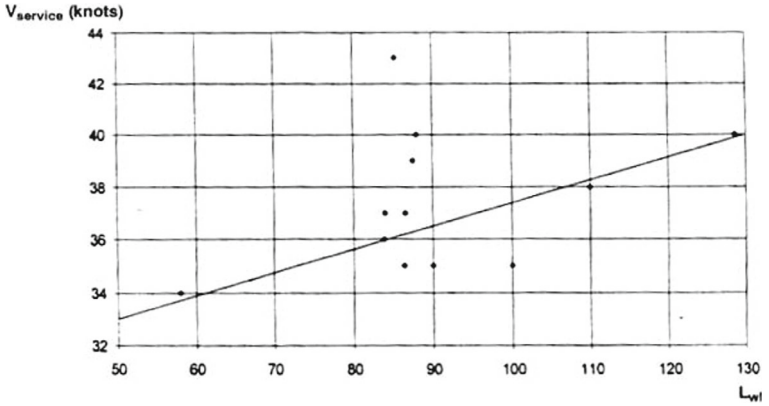


Fig. A.111 Regression analysis of the service speed Vs [knots] versus the length L_{wl} [m] for fast single-hull ferries according to Kristensen (2000) in Friis et al. (2002)

A.11 Car Carrying Catamarans

See Figs. A.112, A.113, A.114 and A.115.

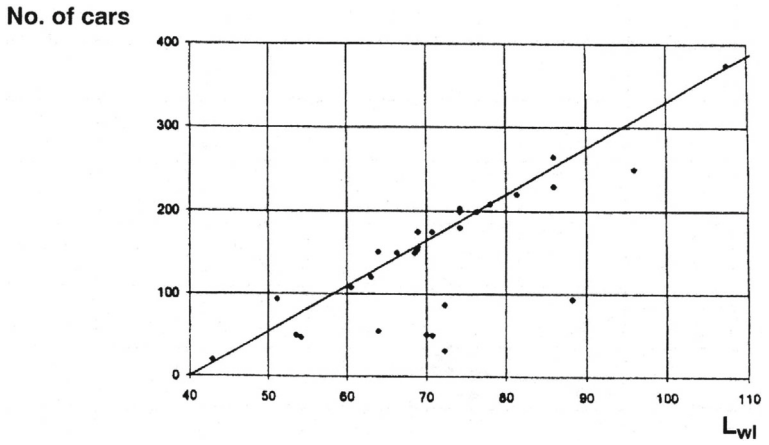


Fig. A.112 Regression analysis of the vehicle number versus the length L_{wl} [m] for catamaran according to Kristensen (2000) in Friis et al. (2002)

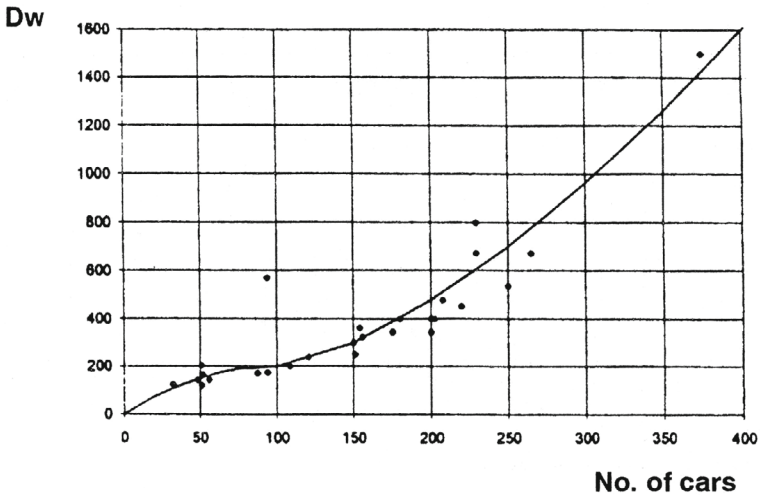


Fig. A.113 Regression analysis of the DWT [tons] versus the vehicle number for catamaran according to Kristensen (2000) in Friis et al. (2002)

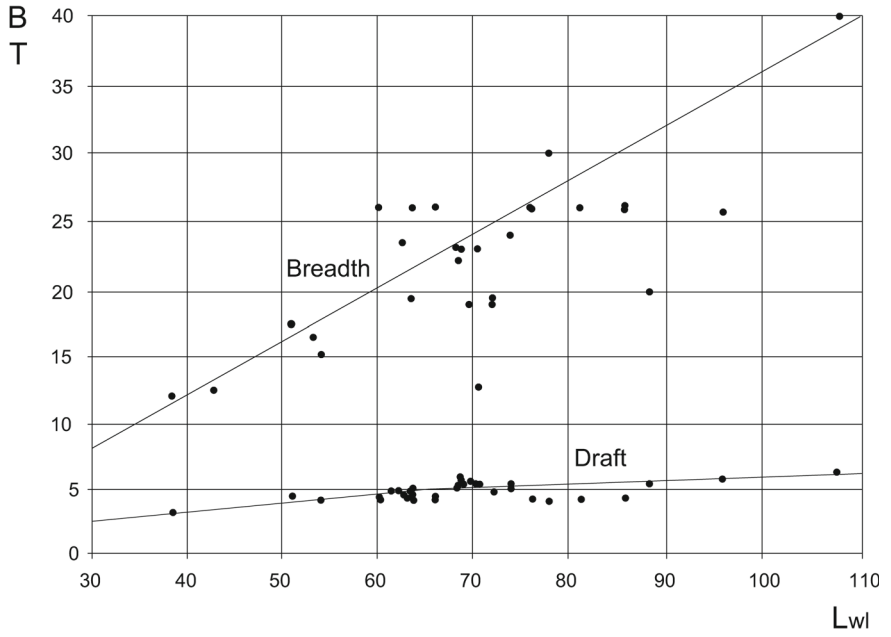


Fig. A.114 Regression analysis of the beam B [m] and draft T [m] versus the length at the waterline L_{wl} [m] for catamaran according to Kristensen (2000) in Friis et al. (2002)

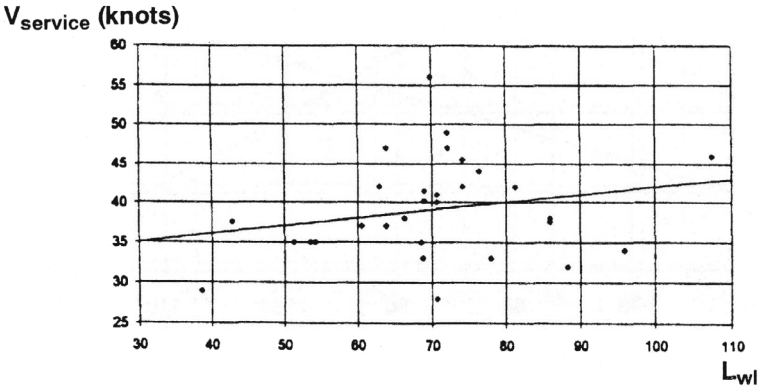


Fig. A.115 Regression analysis of the service speed Vs [knots] versus the length L_{wl} [m] for catamaran ferries according to Kristensen (2000) in Friis et al. (2002)

A.12 Reefer Ships

See Figs. A.116, A.117, A.118, A.119, A.120, A.121, A.122 and A.123.

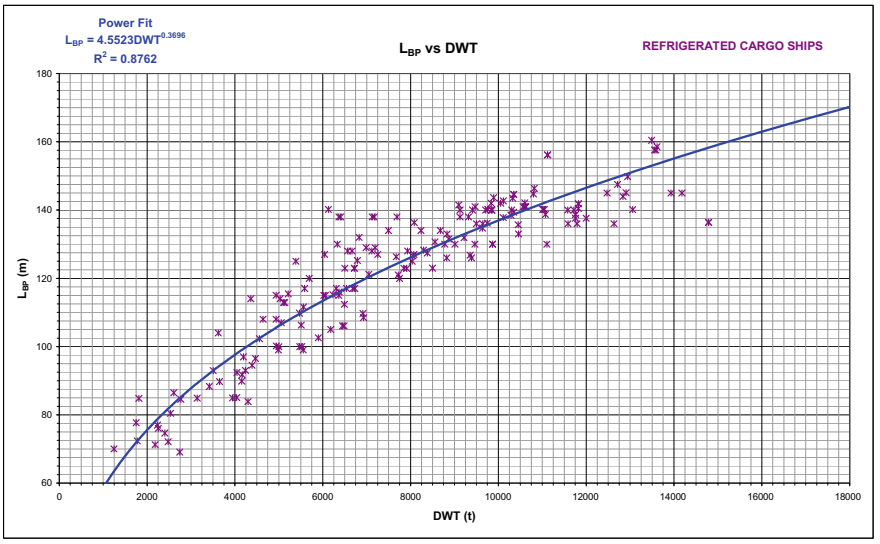


Fig. A.116 Regression analysis of the length L_{BP} [m] versus the DWT [tons] for reefer ships (Kalokairinos et al. 2000–2005)

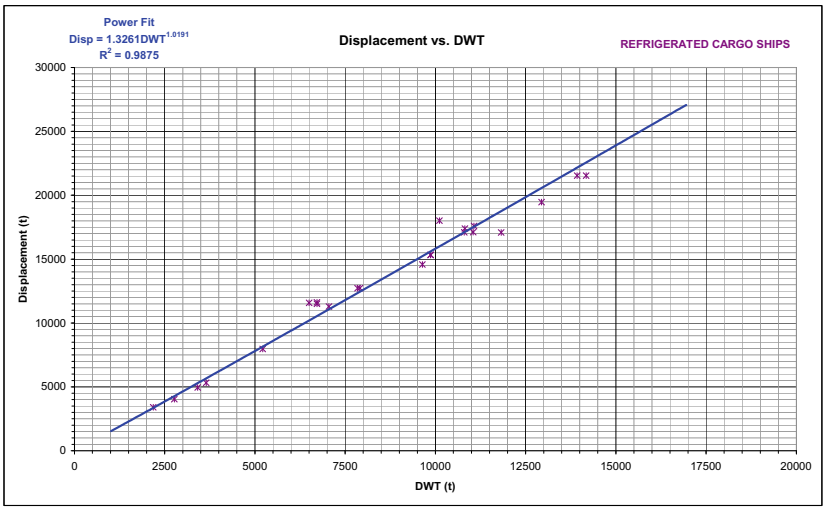


Fig. A.117 Regression analysis of the displacement Δ [tons] versus the DWT [tons] for reefer ships (Kalokairinos et al. 2000–2005)

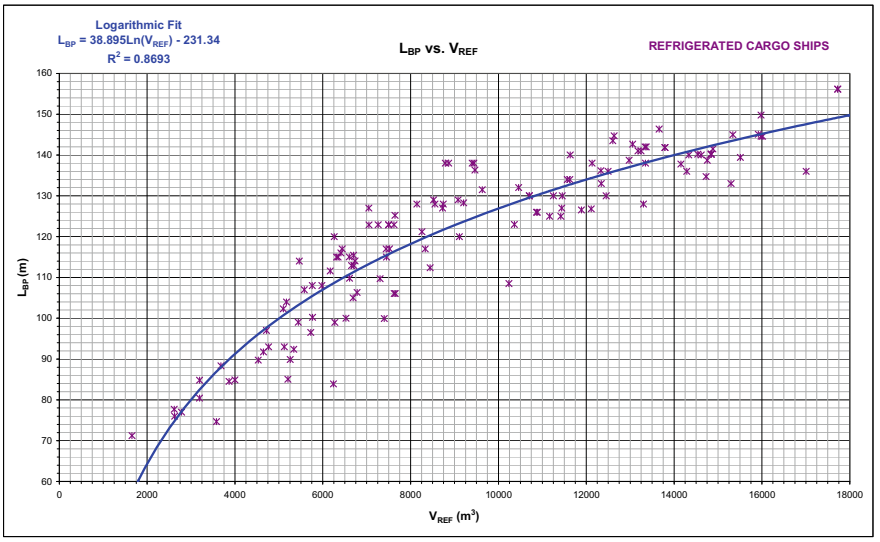


Fig. A.118 Regression analysis of the length L_{BP} [m] versus the refrigerated hold volume V_{REF} [m^3] for reefer ships (Kalokairinos et al. 2000–2005)

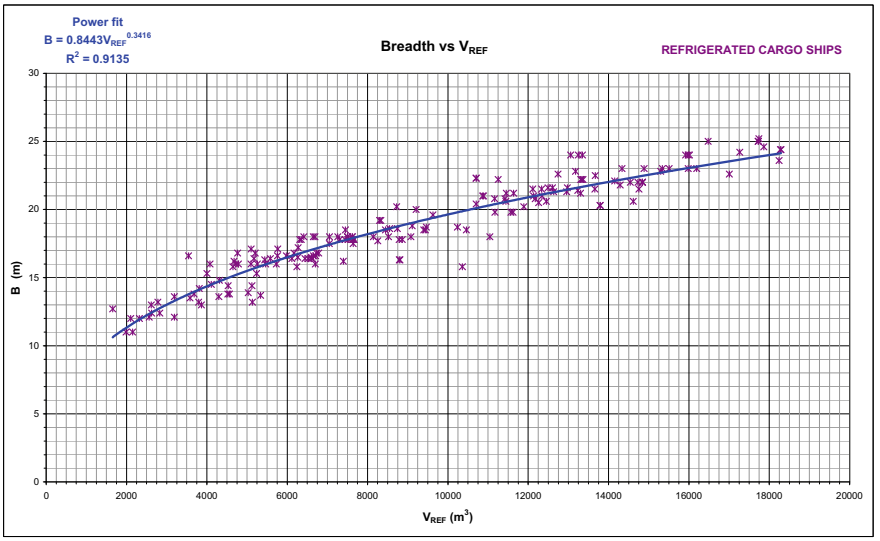


Fig. A.119 Regression analysis of the beam B [m] versus the hold volume VREF [m^3] for reefer ships (Kalokairinos et al. 2000–2005)

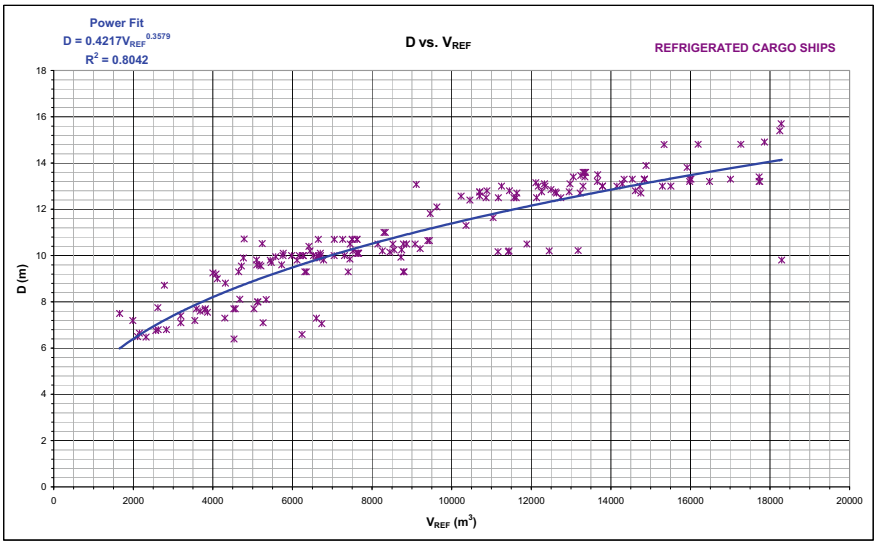


Fig. A.120 Regression analysis of the side depth D [m] versus the hold volume V_{REF} [m^3] for reefer ships (Kalokairinos et al. 2000–2005)

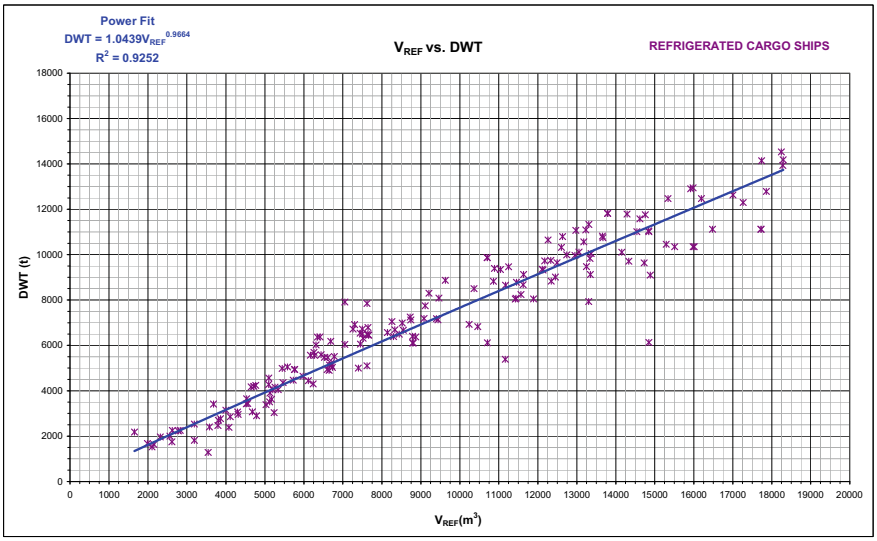


Fig. A.121 Regression analysis of the DWT [tons] versus the hold volume V_{REF} [m^3] for reefer ships (Kalokairinos et al. 2000–2005)

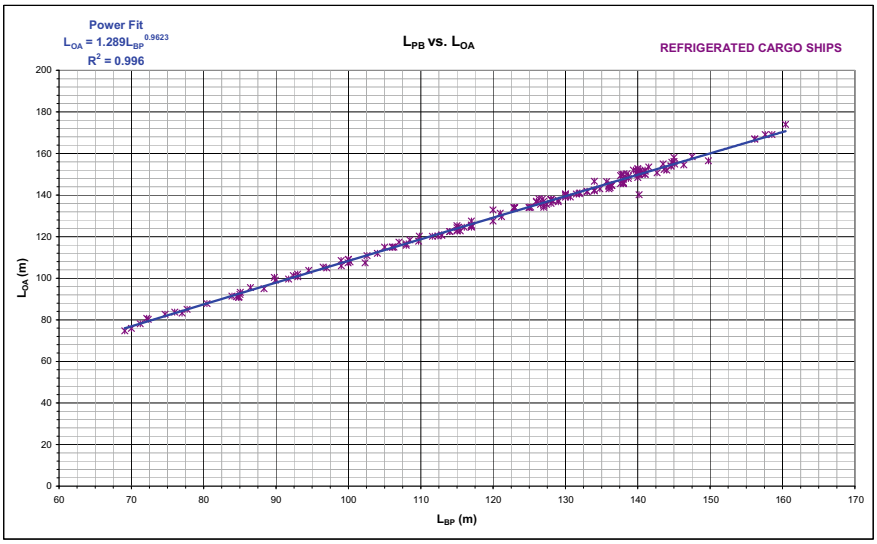


Fig. A.122 Regression analysis of the overall length L_{OA} [m] versus the length between perpendiculars L_{BP} [m] for reefer ships (Kalokairinos et al. 2000–2005)

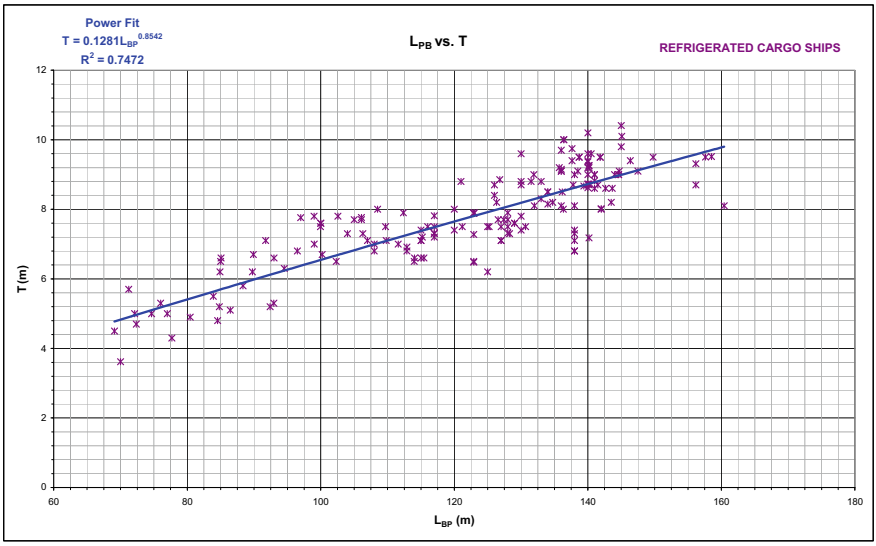


Fig. A.123 Regression analysis of the draft T [m] versus the length between perpendiculars L_{BP} [m] for reefer ships (Kalokairinos et al. 2000–2005)

A.13 Passenger/Cruise Ships

See Figs. A.124, A.125, A.126, A.127, A.128, A.129 and A.130.

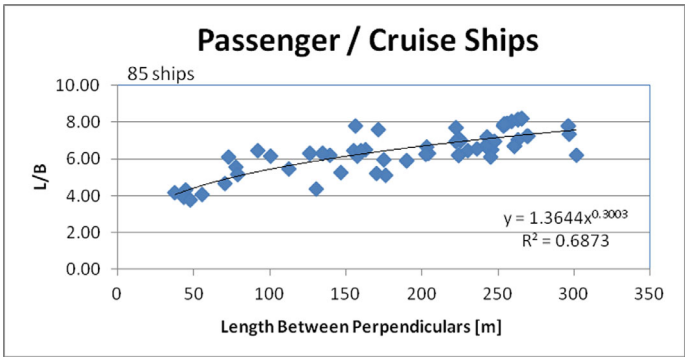


Fig. A.124 Regression analysis of the L/B versus the length between perpendiculars L_{BP} [m] for cruise ships (IHS Fairplay 2011)

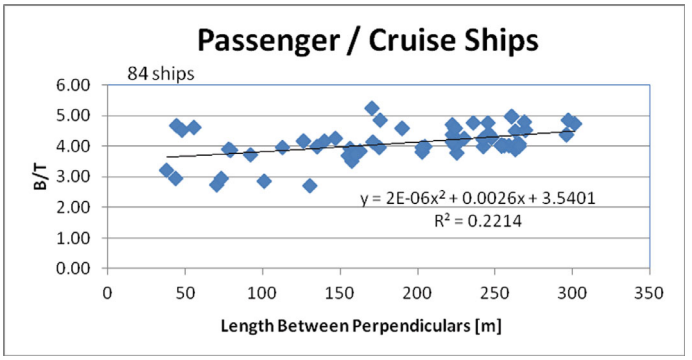


Fig. A.125 Regression analysis of the B/T versus the length between perpendiculars L_{BP} [m] for cruise ships (IHS Fairplay 2011)

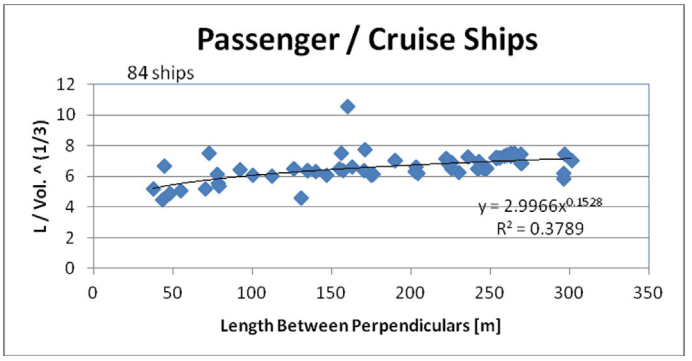


Fig. A.126 Regression analysis of the $L/V^{1/3}$ versus the length between perpendiculars L_{BP} [m] for cruise ships (IHS Fairplay 2011)

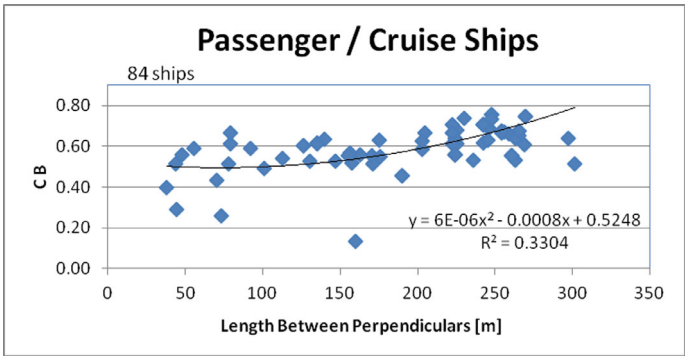


Fig. A.127 Regression analysis of C_B versus the length between perpendiculars L_{BP} [m] for cruise ships (IHS Fairplay 2011)

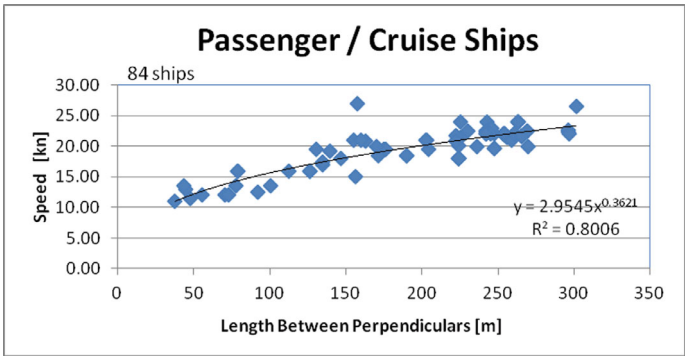


Fig. A.128 Regression analysis of the speed versus the length between perpendiculars L_{BP} [m] for cruise ships (IHS Fairplay 2011)

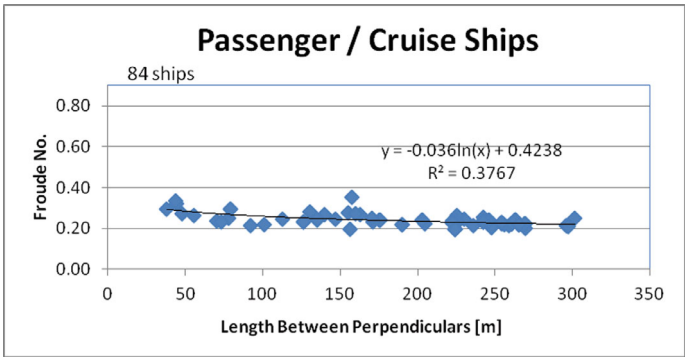


Fig. A.129 Regression analysis of the Froude no. versus the length between perpendiculars L_{BP} [m] for cruise ships (IHS Fairplay 2011)

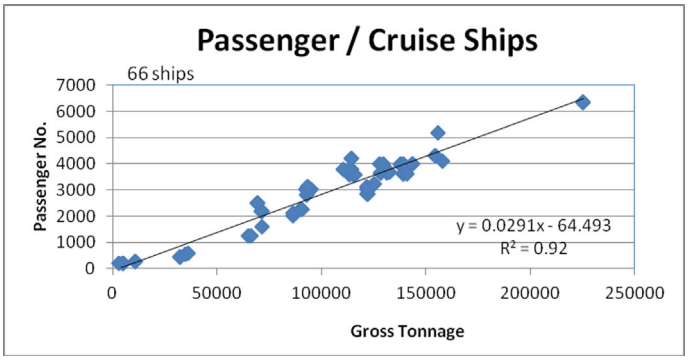


Fig. A.130 Regression analysis of the no. of passengers versus GT for cruise ships

A.14 Offshore Tug/Supply Ships

See Figs. A.131, A.132, A.133, A.134, A.135, A.136, A.137, A.138 and A.139.

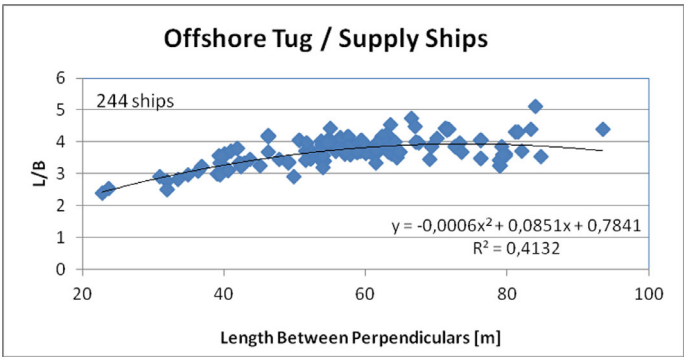


Fig. A.131 Regression analysis of the L/B ratio versus the length between perpendiculars L_{BP} [m] for Offshore Tug/Supply Ships (IHS Fairplay 2011)

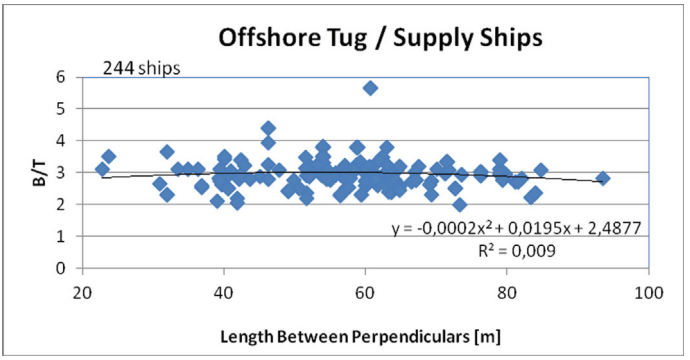


Fig. A.132 Regression analysis of the B/T ratio versus the length between perpendiculars L_{BP} [m] for Offshore Tug/Supply Ships (IHS Fairplay 2011)

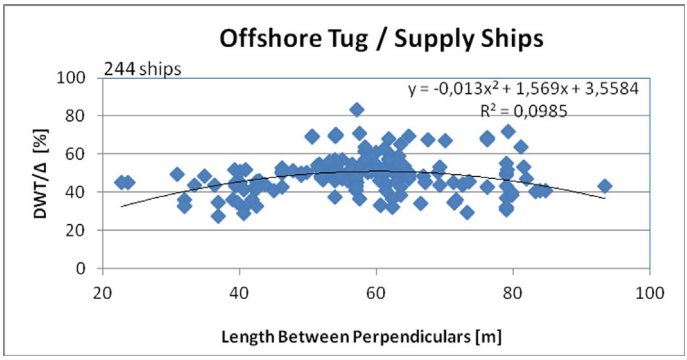


Fig. A.133 Regression analysis of the % DWT/Displacement ratio versus the length between perpendiculars L_{BP} [m] for Offshore Tug/Supply Ships (IHS Fairplay 2011)

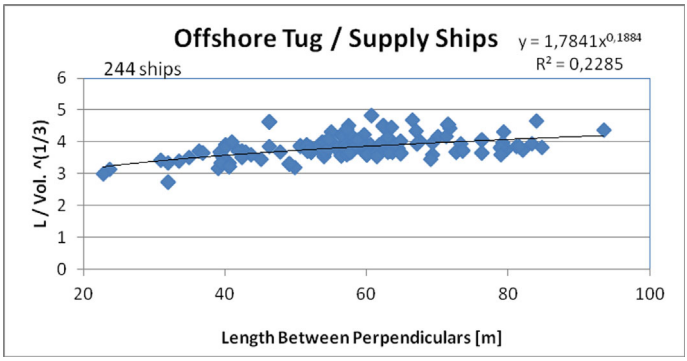


Fig. A.134 Regression analysis of the L/Vol.$^{1/3}$ ratio versus the length between perpendiculars L_{BP} [m] for Offshore Tug/Supply Ships (IHS Fairplay 2011)

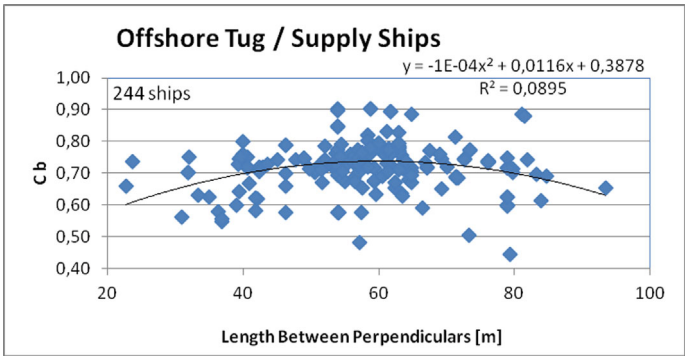


Fig. A.135 Regression analysis of the c_B versus the length between perpendiculars L_{BP} [m] for Offshore Tug/Supply Ships (IHS Fairplay 2011)

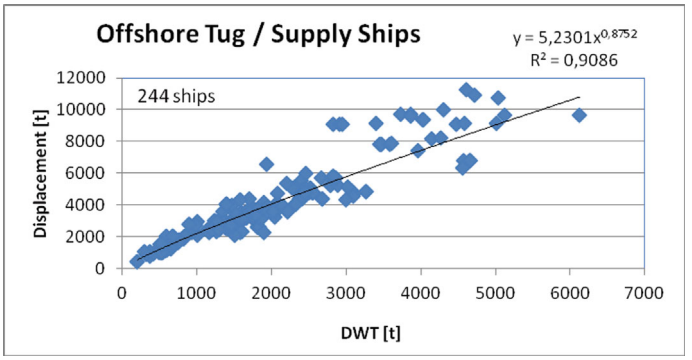


Fig. A.136 Regression analysis of the Displacement [t] versus the DWT [t] for Offshore Tug/ Supply Ships (IHS Fairplay 2011)

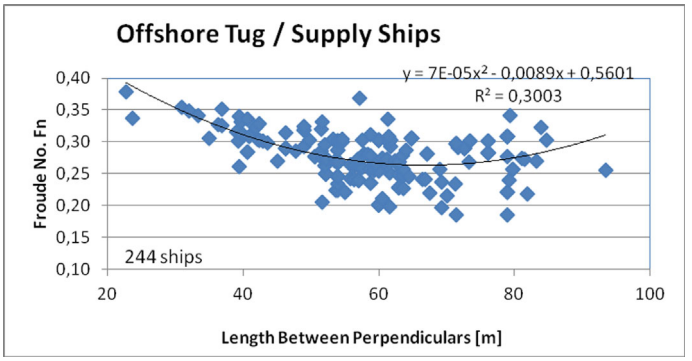


Fig. A.137 Regression analysis of the Froude no. versus the length between perpendiculars L_{BP} [m] for Offshore Tug/Supply Ships (IHS Fairplay 2011)

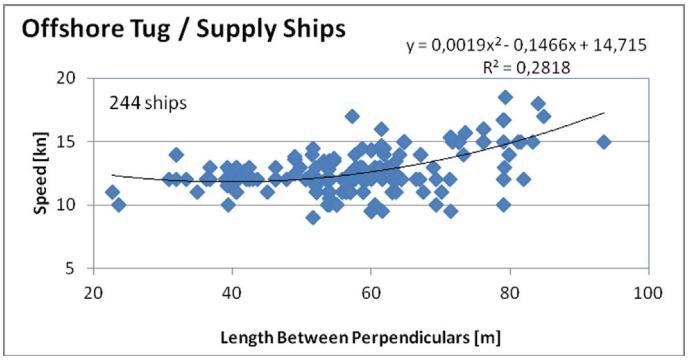


Fig. A.138 Regression analysis of the speed [kn] versus the length between perpendiculars L_{BP} [m] for Offshore Tug/Supply Ships (IHS Fairplay 2011)

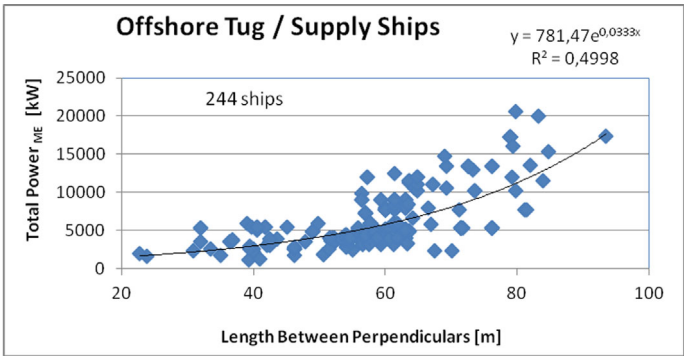


Fig. A.139 Regression analysis of the Main Engine Total Power [kW] versus the length between perpendiculars L_{BP} [m] for Offshore Tug/Supply Ships (IHS Fairplay 2011)

A.15 Fishing Vessels

See Figs. A.140, A.141, A.142, A.143, A.144, A.145, A.146, A.147, A.148, A.149, A.150, A.151, A.152, A.153 and A.154.

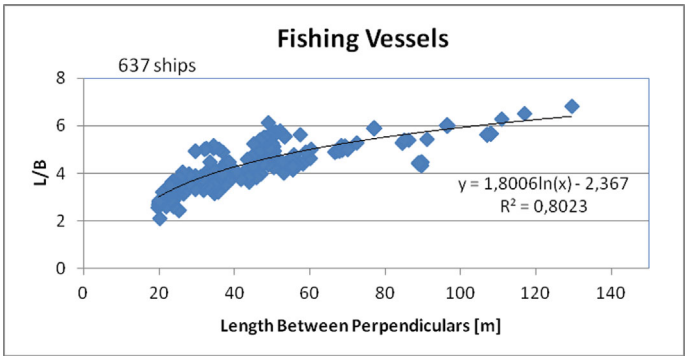


Fig. A.140 Regression analysis of the L/B ratio versus the length between perpendiculars L_{BP} [m] for Fishing Vessels (IHS Fairplay 2011)

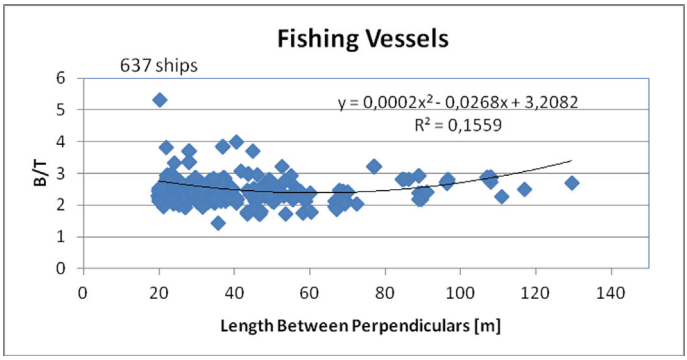


Fig. A.141 Regression analysis of the B/T ratio versus the length between perpendiculars L_{BP} [m] for Fishing Vessels (IHS Fairplay 2011)

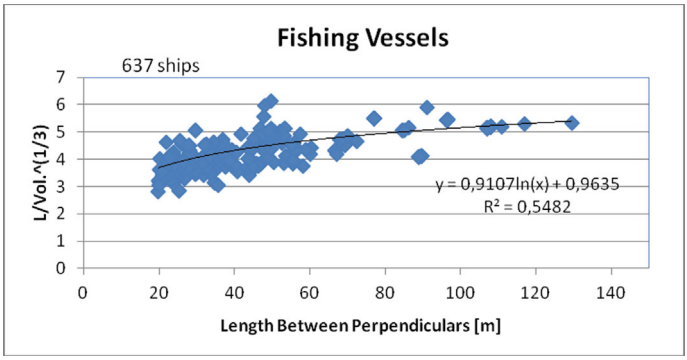


Fig. A.142 Regression analysis of the L/Vol.$^{1/3}$ ratio versus the length between perpendiculars L_{BP} [m] for Fishing Vessels (IHS Fairplay 2011)

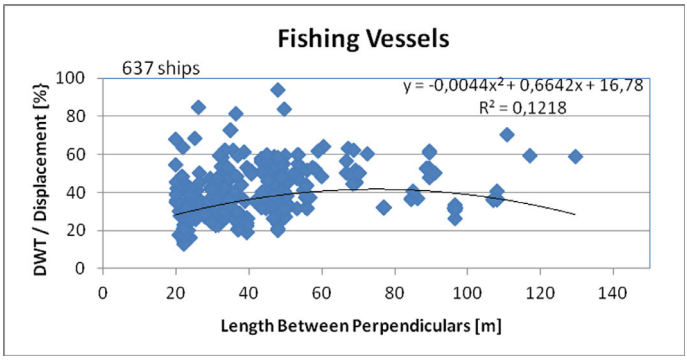


Fig. A.143 Regression analysis of the % DWT/Displacement ratio versus the length between perpendiculars L_{BP} [m] for Fishing Vessels (IHS Fairplay 2011)

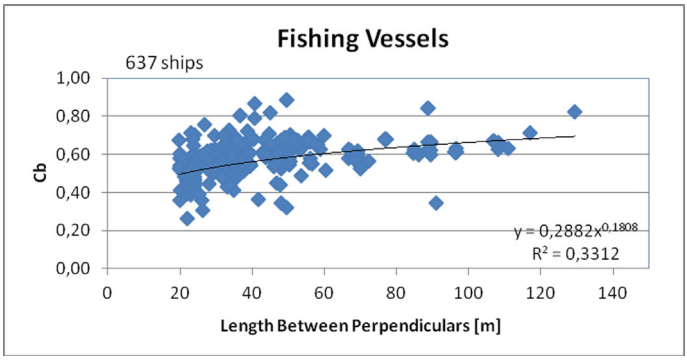


Fig. A.144 Regression analysis of the C_B versus the length between perpendiculars L_{BP} [m] for Fishing Vessels (IHS Fairplay 2011)

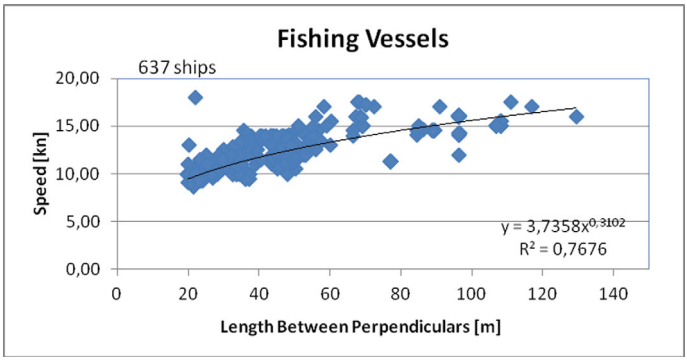


Fig. A.145 Regression analysis of the Speed [kn] versus the length between perpendiculars L_{BP} [m] for Fishing Vessels (IHS Fairplay 2011)

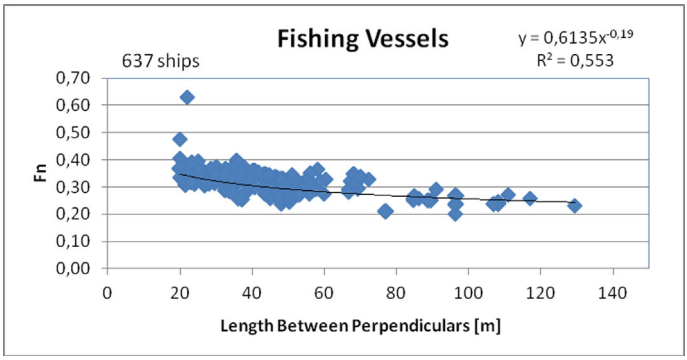


Fig. A.146 Regression analysis of the Froude no. versus the length between perpendiculars L_{BP} [m] for Fishing Vessels (IHS Fairplay 2011)

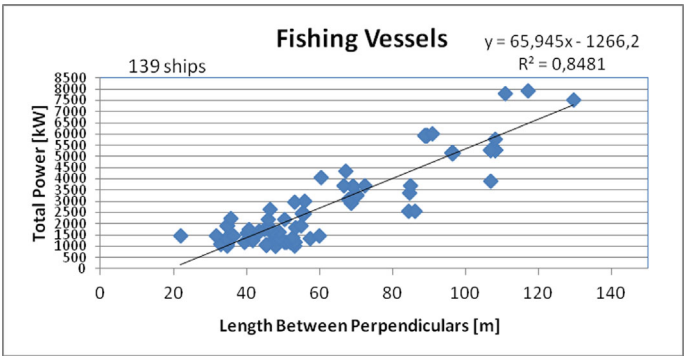


Fig. A.147 Regression analysis of the Main Engine Total Power [kW] versus the length between perpendiculars L_{BP} [m] for Fishing Vessels (IHS Fairplay 2011)

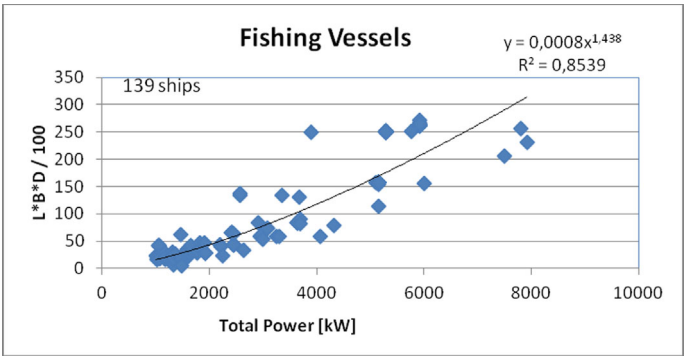


Fig. A.148 Regression analysis of the LBD/100 versus the Main Engine Total Power [kW] for Fishing Vessels (IHS Fairplay 2011)

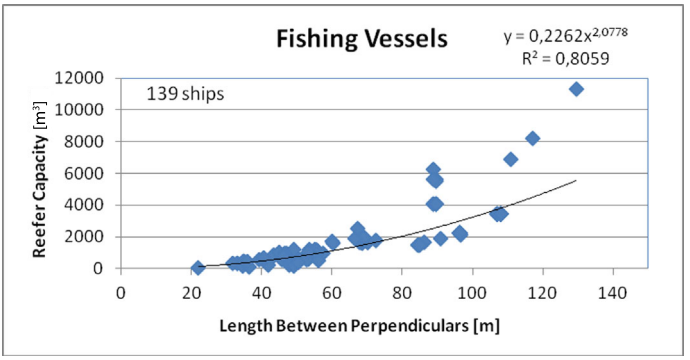


Fig. A.149 Regression analysis of the Reefer Capacity [cubic meters] versus the length between perpendiculars L_{BP} [m] for Fishing Vessels (IHS Fairplay 2011)

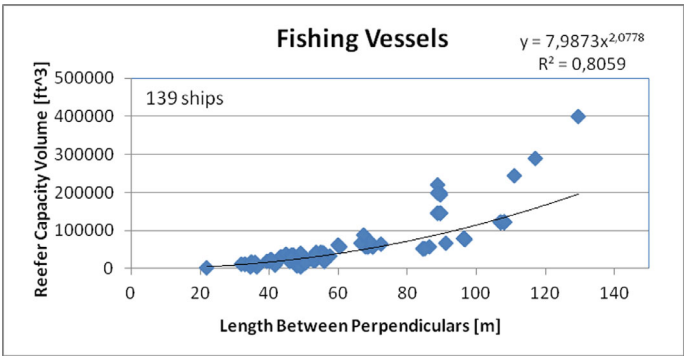


Fig. A.150 Regression analysis of the Reefer Capacity Volume [ft^3] versus the length between perpendiculars L_{BP} [m] for Fishing Vessels (IHS Fairplay 2011)

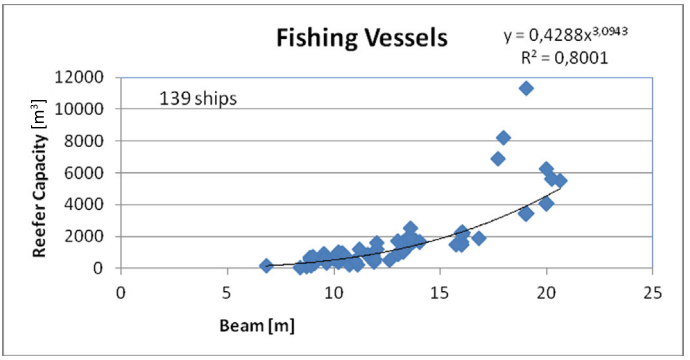


Fig. A.151 Regression analysis of the Reefer Capacity [cubic meters] versus Beam B [m] for Fishing Vessels (IHS Fairplay 2011)

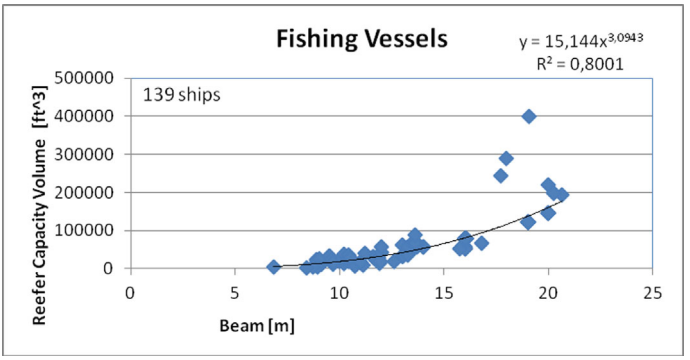


Fig. A.152 Regression analysis of the Reefer Capacity Volume [ft^3] versus the Beam B [m] for Fishing Vessels (IHS Fairplay 2011)

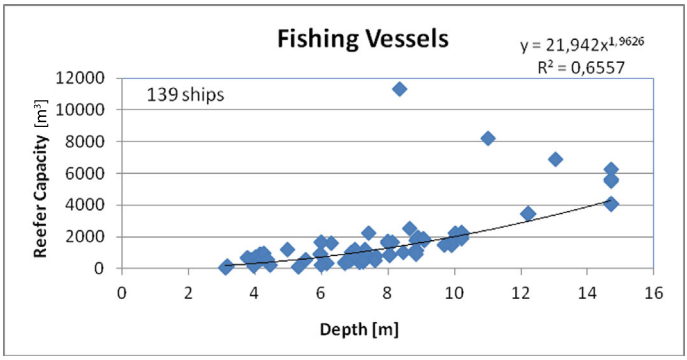


Fig. A.153 Regression analysis of the Reefer Capacity [cubic meters] versus the Depth D [m] for Fishing Vessels (IHS Fairplay 2011)

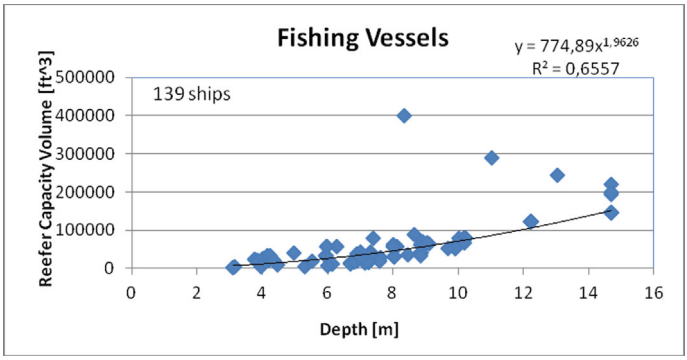


Fig. A.154 Regression analysis of the Reefer Capacity Volume [ft^3] versus the Depth D [m] for Fishing Vessels (IHS Fairplay 2011)

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